2.02 Trigonometry: $\sin x$, $\cos x$, $\tan x$ in Radians

Worked Solutions

Notes on the method used here for solving trigonometric equations:

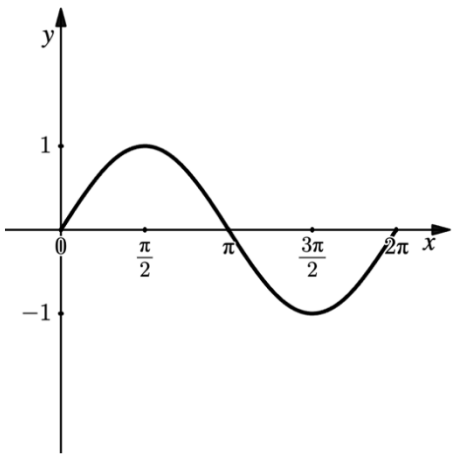
The method used here may in some instances appear like a long-winded way of getting to the required solutions, but this is to construct a general approach that will work for all situations, including more complicated examples.

For sine and cosine equations, we start by finding the two solutions in the "first period" of the function. For example, for $\sin x = 0.3$ this would be the two solutions within $0 \leq x < 2\pi$, and for $\cos 2x = 0.6$ this would be the two solutions within $0 \leq x < \pi$. Then we add and/or subtract the function's period to find all the required solutions.

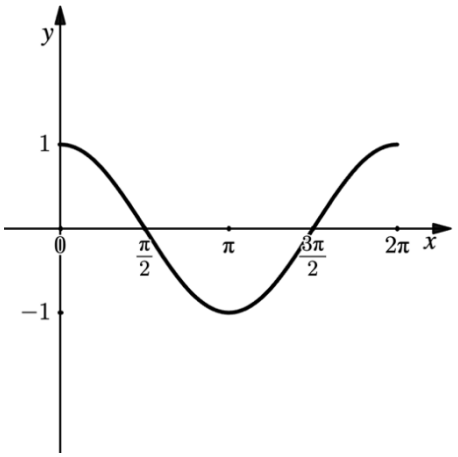
For tangent equations, we start by finding the first solution. For example, for $\tan x = 2$ this would be $x = \arctan 2$, and for $\tan 3x = 6.4$ this would be $x = \frac{1}{3}\arctan 6.4$. Then we add and/or subtract the function's period to find all the required solutions.

The diagrams used in the questions are for illustrative purposes only to help spot symmetries and the horizontal lines do not always reflect the scaling in the question.

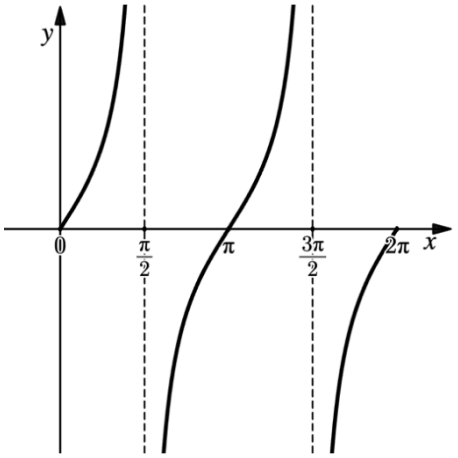
1 a)



1 b)



1 c)



2 a)

$$\sin\theta = 0 \text{ for } 0 \leq \theta < 2\pi$$
$$\Rightarrow \theta = \{0, \pi\}.$$

2 b)

$$\sin\theta = 1 \text{ for } 0 \leq \theta < 2\pi$$
$$\Rightarrow \theta = \{\frac{\pi}{2}\}.$$

2 c)

$$\sin\theta = -1 \text{ for } 0 \leq \theta < 2\pi$$
$$\Rightarrow \theta = \{\frac{3\pi}{2}\}.$$

3 a)

$$\cos\theta = 0 \text{ for } 0 \leq \theta < 2\pi$$
$$\Rightarrow \theta = \{\frac{\pi}{2}, \frac{3\pi}{2}\}.$$

3 b)

$$\cos\theta = 1 \text{ for } 0 \leq \theta < 2\pi$$
$$\Rightarrow \theta = \{0\}.$$

3 c)

$$\cos\theta = -1 \text{ for } 0 \leq \theta < 2\pi$$
$$\Rightarrow \theta = \{\pi\}.$$

4 a)

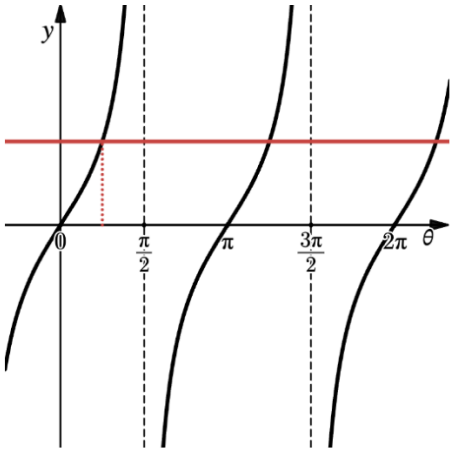
$$\tan\theta = 0 \text{ for } 0 \leq \theta < 2\pi$$
$$\Rightarrow \theta = \{0, \pi\}.$$

4 b)

$\tan\theta = 1$ for $0 \leq \theta < 2\pi$.

Find first value:

$$\theta = \arctan 1 = \frac{\pi}{4} \quad \text{(in range)}.$$



Add period:

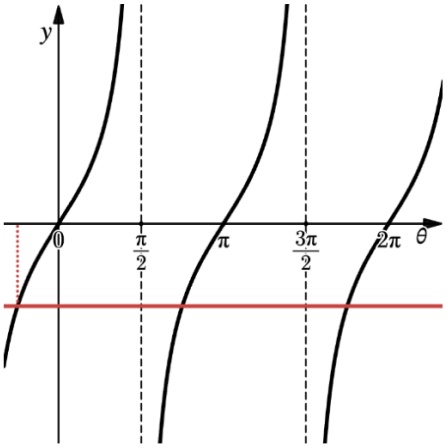
$$\theta = \frac{\pi}{4} + \pi = \frac{5\pi}{4} \quad \text{(in range)}$$
$$\Rightarrow \theta = \{\frac{\pi}{4}, \frac{5\pi}{4}\}.$$

4 c)

$\tan\theta = -1$ for $0 \leq \theta < 2\pi$.

Find first value:

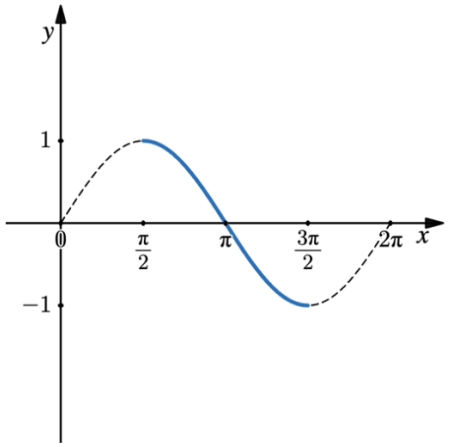
$$\theta = \arctan(-1) = -\frac{\pi}{4} \quad \text{(not in range)}.$$



Add on period:

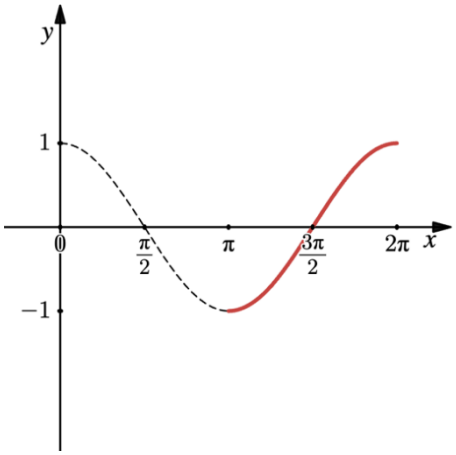
$$\theta = -\frac{\pi}{4} + \pi = \frac{3\pi}{4} \quad \text{(in range)}$$
$$\theta = \frac{3\pi}{4} + \pi = \frac{7\pi}{4} \quad \text{(in range)}$$
$$\Rightarrow \theta = \{\frac{3\pi}{4}, \frac{7\pi}{4}\}.$$

5)



$$\frac{\pi}{2} < x < \frac{3\pi}{2}.$$

6)



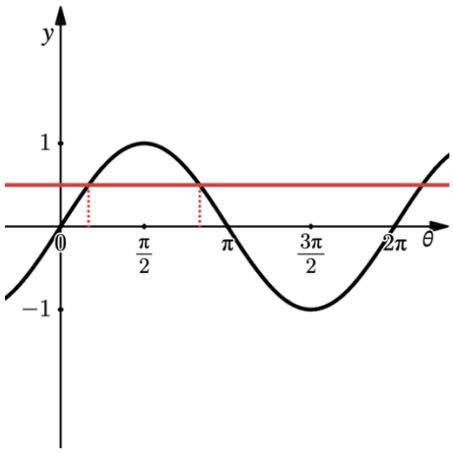
$$\pi < x < 2\pi.$$

7 a)

$\sin\theta = \frac{1}{4}$ for $0 \leq \theta < 2\pi$.

Find first value between 0 and 2π:

$$\theta = \arcsin\frac{1}{4} = 0.2526\ldots \quad \text{(in range)}.$$



Find second value between 0 and 2π:

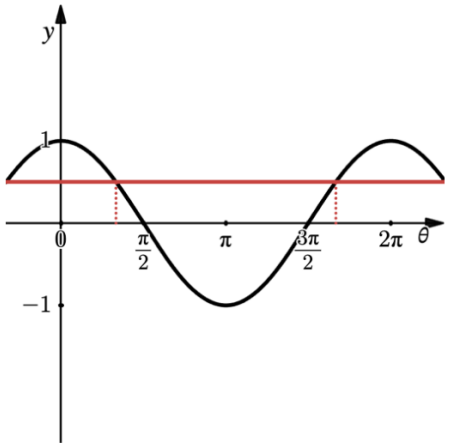
$$\theta = \pi - 0.2526\ldots = 2.8889\ldots \quad \text{(in range)}$$
$$\theta = \{0.253, 2.89\} \text{ to 3 s.f.}$$

7 b)

$\cos\theta = \frac{3}{4}$ for $0 \leq \theta < 2\pi$.

Find first value between 0 and 2π:

$$\theta = \arccos\frac{3}{4} = 0.7227\ldots \quad \text{(in range)}.$$



Find second value between 0 and 2π:

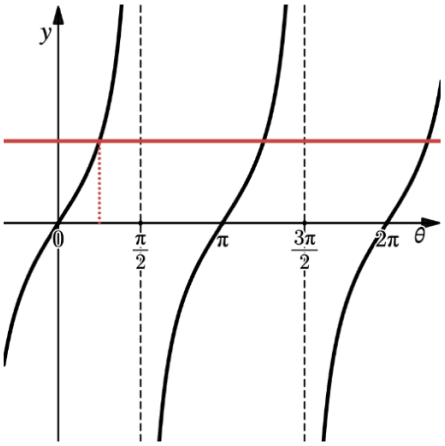
$$\theta = 2\pi - 0.7227\ldots = 5.5604\ldots \quad \text{(in range)}$$
$$\Rightarrow \theta = \{0.723, 5.56\} \text{ to 3 s.f.}$$

7 c)

$\tan\theta = \frac{5}{4}$ for $0 \leq \theta < 2\pi$.

Find first value:

$$\theta = \arctan\frac{5}{4} = 0.8960\ldots \quad \text{(in range)}.$$



Add period:

$$\theta = 0.8960\ldots + \pi = 4.0376\ldots \quad \text{(in range)}$$

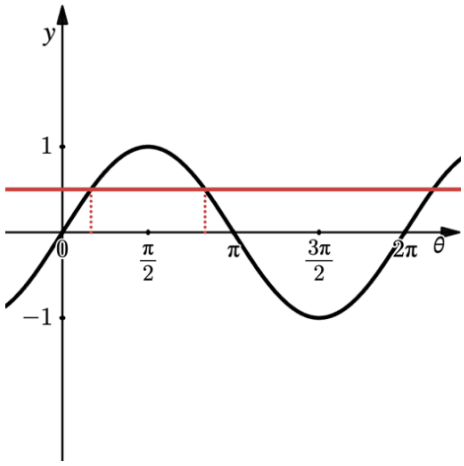
$$\theta = \{0.896, 4.04\} \text{ to 3 s.f.}$$

8 a)

$\sin\theta = 0.9$ for $-2\pi \leq \theta < 0$.

Find first value between 0 and 2π:

$$\theta = \arcsin 0.9 = 1.1197\ldots \quad \text{(not in range)}.$$



Find second value between 0 and 2π:

$$\theta = \pi - 1.1197\ldots = 2.0218\ldots \quad \text{(not in range)}$$

Subtract period:

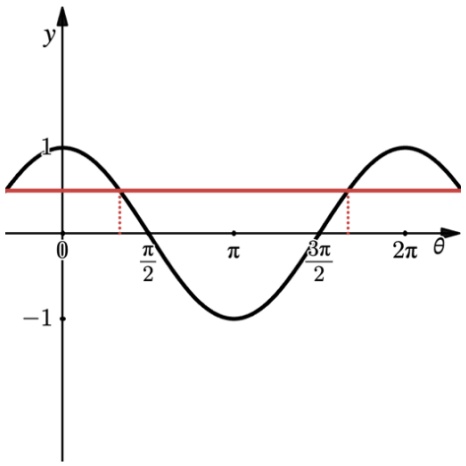
$$\theta = 1.1197\ldots - 2\pi = -5.1634\ldots \quad \text{(in range)}$$
$$\theta = 2.0218\ldots - 2\pi = -4.2613\ldots \quad \text{(in range)}$$
$$\theta = \{-5.16, -4.26\} \text{ to 3 s.f.}$$

8 b)

$\cos\theta = \frac{1}{9}$ for $-2\pi \leq \theta < 0$.

Find first value between 0 and 2π:

$$\theta = \arccos\frac{1}{9} = 1.4594\ldots \quad \text{(not in range)}.$$



Find second value between 0 and 2π:

$$\theta = 2\pi - 1.4594\ldots = 4.8237\ldots \quad \text{(not in range)}$$

Subtract period:

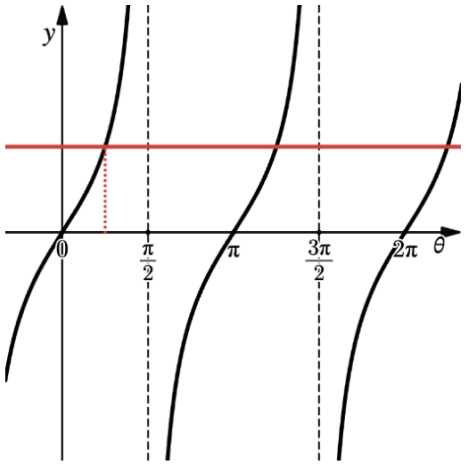
$$\theta = 1.4594\ldots - 2\pi = -4.8237\ldots \quad \text{(in range)}$$
$$\theta = 4.8237\ldots - 2\pi = -1.4594\ldots \quad \text{(in range)}$$
$$\Rightarrow \theta = \{-4.82, -1.46\} \text{ to 3 s.f.}$$

8 c)

$\tan\theta = \frac{6}{5}$ for $-2\pi \leq \theta < 0$.

Find first value:

$$\theta = \arctan\frac{6}{5} = 0.8760\ldots \quad \text{(not in range)}.$$



Subtract period:

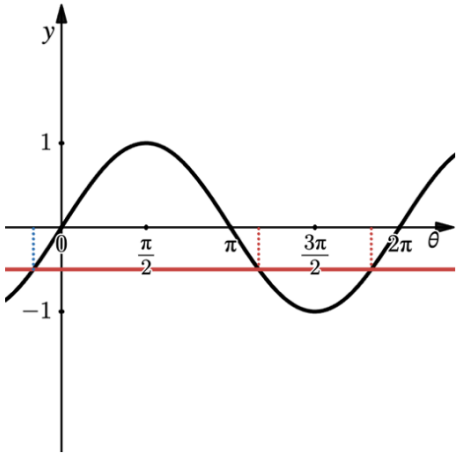
$$\theta = 0.8760\ldots - \pi = -2.2655\ldots \quad \text{(in range)}$$
$$\theta = -2.2655\ldots - \pi = -5.4071\ldots \quad \text{(in range)}$$
$$\theta = \{-5.41, -2.27\} \text{ to 3 s.f.}$$

9 a)

$\sin\theta = -\frac{1}{6}$ for $0 \leq \theta < 2\pi$.

Find first value:

$$\theta = \arcsin(-\frac{1}{6}) = -0.1674\ldots \quad \text{(not in range)}.$$



Find first value between 0 and 2π:

$$\theta = \pi + 0.1674\ldots = 3.3090\ldots \quad \text{(in range)}$$

Find second value between 0 and 2π:

$$\theta = 2\pi - 0.1674\ldots = 6.1157\ldots \quad \text{(in range)}$$

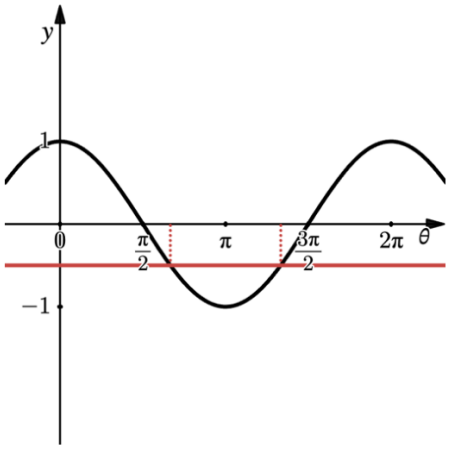
$$\theta = \{3.31, 6.12\} \text{ to 3 s.f.}$$

9 b)

$\cos\theta = -\frac{8}{9}$ for $0 \leq \theta < 2\pi$.

Find first value between 0 and 2π:

$$\theta = \arccos(-\frac{8}{9}) = 2.6657\ldots \quad \text{(in range)}.$$



Find second value between 0 and 2π:

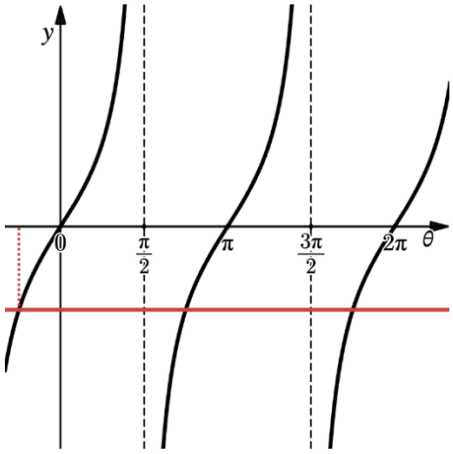
$$\theta = 2\pi - 2.6657\ldots = 3.6174\ldots \quad \text{(in range)}$$
$$\Rightarrow \theta = \{2.67, 3.62\} \text{ to 3 s.f.}$$

9 c)

$\tan\theta = -\frac{1}{3}$ for $0 \leq \theta < 2\pi$.

Find first value:

$$\theta = \arctan(-\frac{1}{3}) = -0.3217\ldots \quad \text{(not in range)}.$$



Add on period:

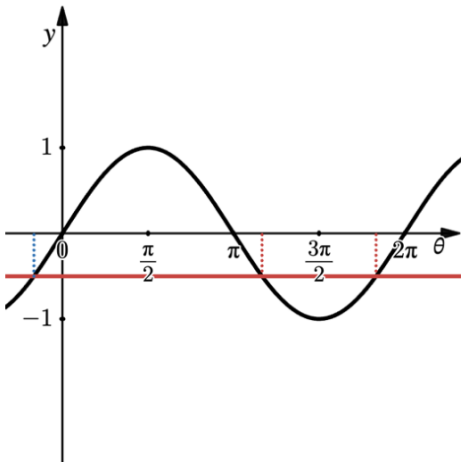
$$\theta = -0.3217\ldots + \pi = 2.8198\ldots \quad \text{(in range)}$$
$$\theta = 2.8198\ldots + \pi = 5.9614\ldots \quad \text{(in range)}$$
$$\Rightarrow \theta = \{2.82, 5.96\} \text{ to 3 s.f.}$$

10 a)

$\sin\theta = -0.66$ for $-2\pi \leq \theta < 0$.

Find first value:

$$\theta = \arcsin(-0.66) = -0.7208\ldots \quad \text{(in range)}.$$



Find first value between 0° and 360°:

$$\theta = \pi + 0.7208\ldots = 3.8624\ldots \quad \text{(not in range)}$$

Find second value between 0 and 360:

$$\theta = 2\pi - 0.7208\ldots = 5.5623\ldots \quad \text{(not in range)}$$

Subtract period:

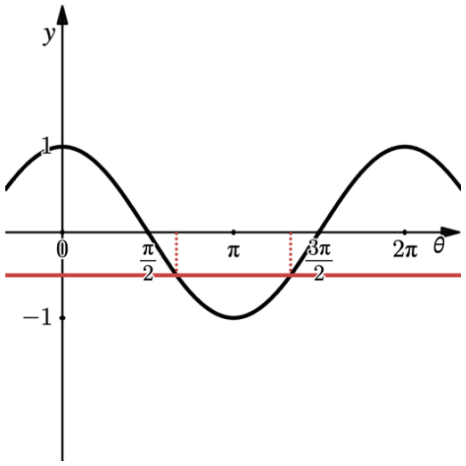
$$\theta = 3.8624\ldots - 2\pi = -2.4207\ldots \quad \text{(in range)}$$
$$\theta = 5.5623\ldots - 2\pi = -0.7208\ldots \quad \text{(in range)}$$
$$\theta = \{-2.42, -0.721\} \text{ to 3 s.f.}$$

10 b)

$\cos\theta = -0.55$ for $-2\pi \leq \theta < 0$.

Find first value between 0 and 2π:

$$\theta = \arccos(-0.55) = 2.1531\ldots \quad \text{(not in range).}$$



Find second value between 0 and 2π:

$$\theta = 2\pi - 2.1531\ldots = 4.1300\ldots \quad \text{(not in range)}$$

Subtract period:

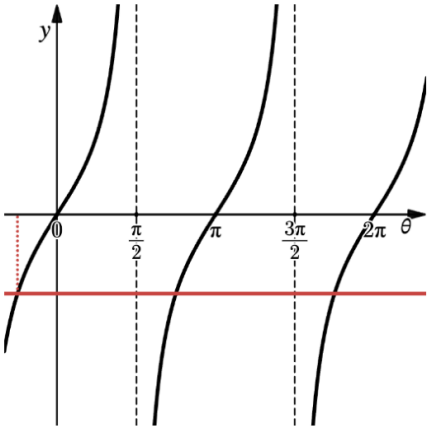
$$\theta = 2.1531\ldots - 2\pi = -4.1300\ldots \quad \text{(in range)}$$
$$\theta = 4.1300\ldots - 2\pi = -2.1531\ldots \quad \text{(in range)}$$
$$\Rightarrow \theta = \{-4.13, -2.15\} \text{ to 3 s.f.}$$

10 c)

$\tan\theta = -8.6$ for $-2\pi \leq \theta < 0$.

Find first value:

$$\theta = \arctan(-8.6) = -1.4550\ldots \quad \text{(in range)}.$$



Subtract period:

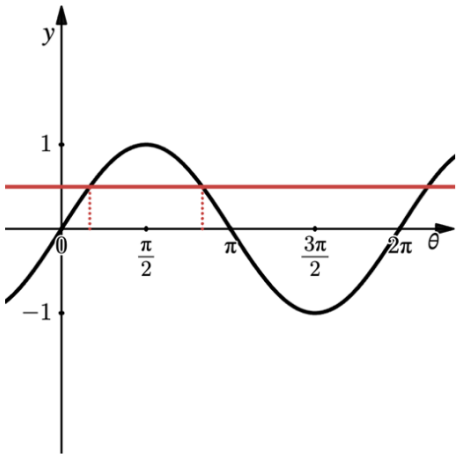
$$\theta = -1.4550\ldots - \pi = -4.5966\ldots \quad \text{(in range)}$$
$$\Rightarrow \theta = \{-4.60, -1.46\} \text{ to 3 s.f.}$$

11 a)

$\sin\theta = \frac{4}{9}$ for $-2\pi \leq \theta < 2\pi$.

Find first value between 0 and 2π:

$$\theta = \arcsin\frac{4}{9} = 0.4605\ldots \quad \text{(in range)}.$$



Find second value between 0 and 2π:

$$\theta = \pi - 0.4605\ldots = 2.6810\ldots \quad \text{(in range)}$$

Subtract period:

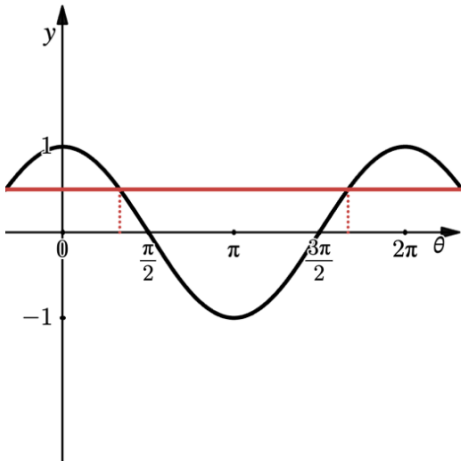
$$\theta = 0.4605\ldots - 2\pi = -5.8226\ldots \quad \text{(in range)}$$
$$\theta = 2.6810\ldots - 2\pi = -3.6021\ldots \quad \text{(in range)}$$
$$\theta = \{-5.82, -3.60, 0.461, 2.68\} \text{ to 3 s.f.}$$

11 b)

$\cos\theta = 0.11$ for $-2\pi \leq \theta < 2\pi$.

Find first value between 0 and 2π:

$$\theta = \arccos 0.11 = 1.4605\ldots \quad \text{(in range)}.$$



Find second value between 0 and 2π:

$$\theta = 2\pi - 1.4605\ldots = 4.8226\ldots \quad \text{(in range)}$$

Subtract period:

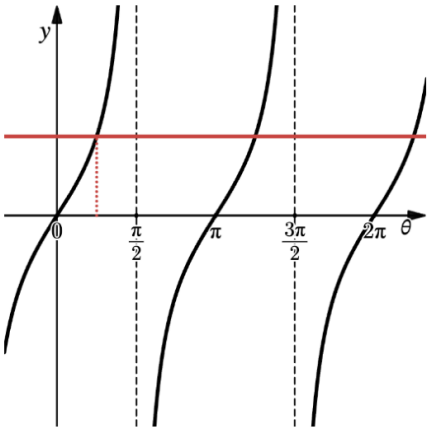
$$\theta = 1.4605\ldots - 2\pi = -4.8226\ldots \quad \text{(in range)}$$
$$\theta = 4.8226\ldots - 2\pi = -1.4605\ldots \quad \text{(in range)}$$
$$\Rightarrow \theta = \{-4.82, -1.46, 1.46, 4.82\} \text{ to 3 s.f.}$$

11 c)

$\tan\theta = 3.2$ for $-2\pi \leq \theta < 2\pi$.

Find first value:

$$\theta = \arctan 3.2 = 1.2679\ldots \quad \text{(in range)}.$$



Add period:

$$\theta = 1.2679\ldots + \pi = 4.4095\ldots \quad \text{(in range)}$$

Subtract period:

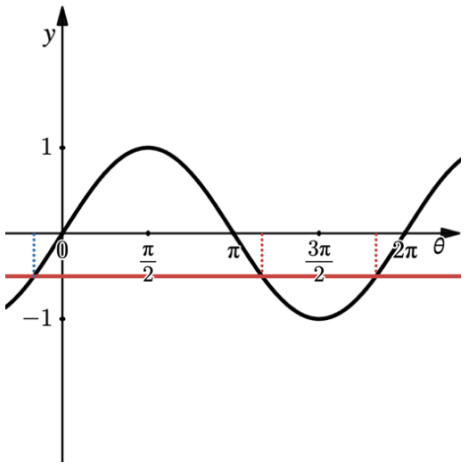
$$\theta = 1.2679\ldots - \pi = -1.8736\ldots \quad \text{(in range)}$$
$$\theta = -1.8736\ldots - \pi = -5.0152\ldots \quad \text{(in range)}$$
$$\theta = \{-5.02, -1.87, 1.27, 4.41\} \text{ to 3 s.f.}$$

11 d)

$\sin\theta = -0.83$ for $-2\pi \leq \theta < 2\pi$.

Find first value:

$$\theta = \arcsin(-0.83) = -0.9791\ldots \quad \text{(in range)}.$$



Find first value between 0 and 2π:

$$\theta = \pi + 0.9791\ldots = 4.1207\ldots \quad \text{(in range)}$$

Find second value between 0 and 2π:

$$\theta = 2\pi - 0.9791\ldots = 5.3040\ldots^\circ \quad \text{(in range)}$$

Subtract period:

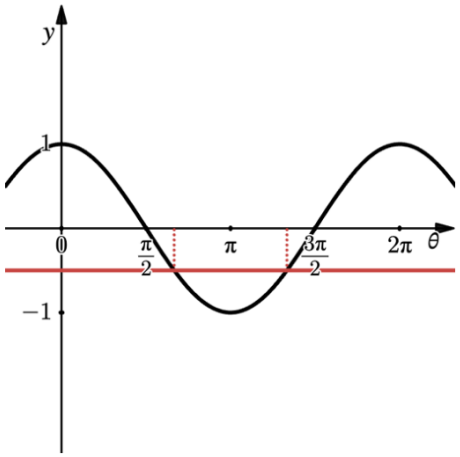
$$\theta = 4.1207\ldots - 2\pi = -2.1624\ldots \quad \text{(in range)}$$
$$\theta = 5.3040\ldots - 2\pi = -0.9791\ldots \quad \text{(in range)}$$
$$\theta = \{-2.16, -0.979, 4.12, 5.30\} \text{ to 3 s.f.}$$

11 e)

$\cos\theta = -\frac{10}{99}$ for $-2\pi \leq \theta < 2\pi$.

Find first value between 0 and 2π:

$$\theta = \arccos(-\frac{10}{99}) = 1.6719\ldots \quad \text{(in range)}.$$



Find second value between 0 and 2π:

$$\theta = 2\pi - 1.6719\ldots = 4.6112\ldots \quad \text{(in range)}$$

Subtract period:

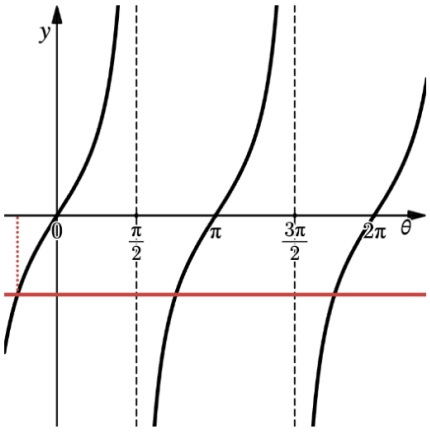
$$\theta = 1.6719\ldots - 2\pi = -4.6112\ldots \quad \text{(in range)}$$
$$\theta = 4.6112\ldots - 2\pi = -1.6719\ldots \quad \text{(in range)}$$
$$\Rightarrow \theta = \{-4.61, -1.67, 1.67, 4.61\} \text{ to 3 s.f.}$$

11 f)

$\tan\theta = -6.6$ for $-2\pi \leq \theta < 2\pi$.

Find first value:

$$\theta = \arctan(-6.6) = -1.4204\ldots \quad \text{(in range)}.$$



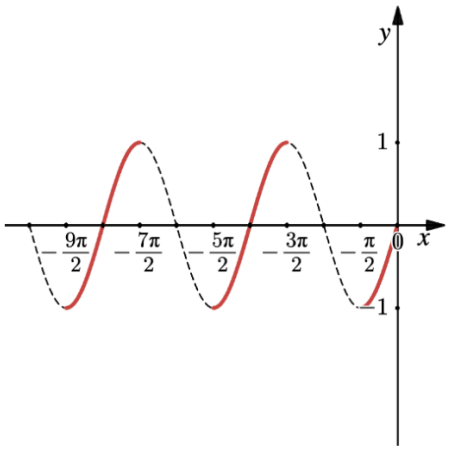
Add period:

$$\theta = -1.4204\ldots + \pi = 1.7211\ldots \quad \text{(in range)}$$
$$\theta = 1.7211\ldots + \pi = 4.8627\ldots \quad \text{(in range)}$$

Subtract period:

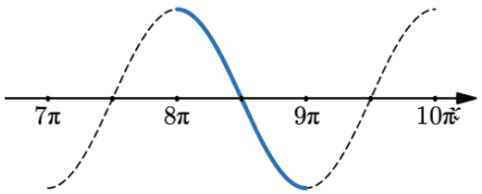
$$\theta = -1.4204\ldots - \pi = -4.5620\ldots \quad \text{(in range)}$$
$$\Rightarrow \theta = \{-4.56, -1.42, 1.72, 4.86\} \text{ to 3 s.f.}$$

12)



$$-\frac{9\pi}{2} < x < -\frac{7\pi}{2} \text{ or } -\frac{5\pi}{2} < x < -\frac{3\pi}{2} \text{ or } -\frac{\pi}{2} < x < 0.$$

13)



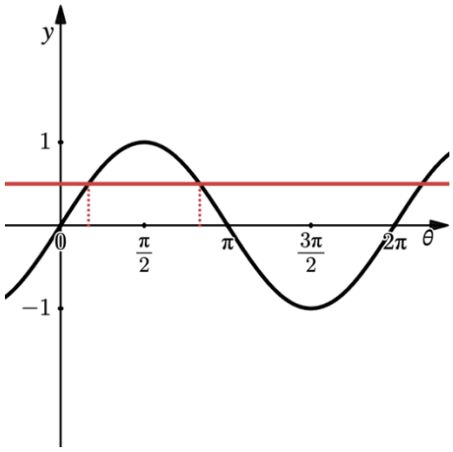
$$8\pi < x < 9\pi.$$

14 a)

$\sin\theta = \frac{3}{16}$ for $0 \leq \theta < 4\pi$.

Find first value between 0 and 2π:

$$\theta = \arcsin\frac{3}{16} = 0.1886\ldots \quad \text{(in range)}.$$



Find second value between 0 and 2π:

$$\theta = \pi - 0.1886\ldots = 2.9529\ldots \quad \text{(in range)}$$

Add period:

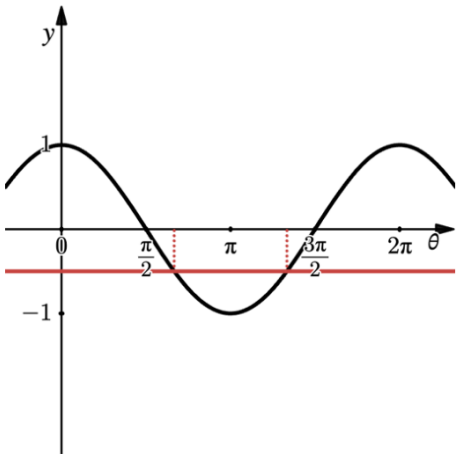
$$\theta = 0.1886\ldots + 2\pi = 6.4718\ldots \quad \text{(in range)}$$
$$\theta = 2.9529\ldots + 2\pi = 9.2361\ldots \quad \text{(in range)}$$
$$\theta = \{0.189, 2.95, 6.47, 9.24\} \text{ to 3 s.f.}$$

14 b)

$\cos\theta = -\frac{5}{11}$ for $3\pi \leq \theta < 6\pi$.

Find first value between 0 and 2π:

$$\theta = \arccos(-\frac{5}{11}) = 2.0426\ldots \quad \text{(not in range)}.$$



Find second value between 0 and 2π:

$$\theta = 2\pi - 2.0426\ldots = 4.2405\ldots \quad \text{(not in range)}$$

Add period:

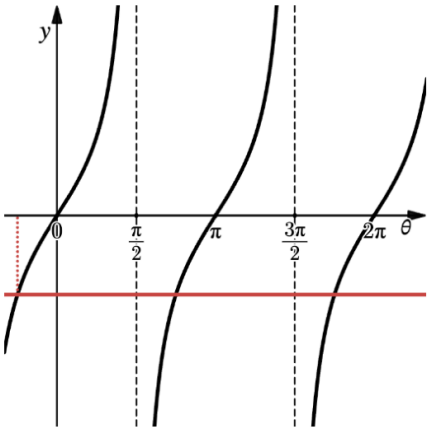
$$\theta = 2.0426\ldots + 2\pi = 8.3258\ldots \quad \text{(not in range)}$$
$$\theta = 8.3258\ldots + 2\pi = 14.6090\ldots \quad \text{(in range)}$$
$$\theta = 4.2405\ldots + 2\pi = 10.5237\ldots \quad \text{(in range)}$$
$$\theta = 10.5237\ldots + 2\pi = 16.8068\ldots \quad \text{(in range)}$$
$$\Rightarrow \theta = \{10.5, 14.6, 16.8\} \text{ to 3 s.f.}$$

14 c)

$\tan\theta = -0.94$ for $-7\pi \leq \theta < -5\pi$.

Find first value:

$$\theta = \arctan(-0.94) = -0.7544\ldots \quad \text{(not in range)}.$$



Subtract period:

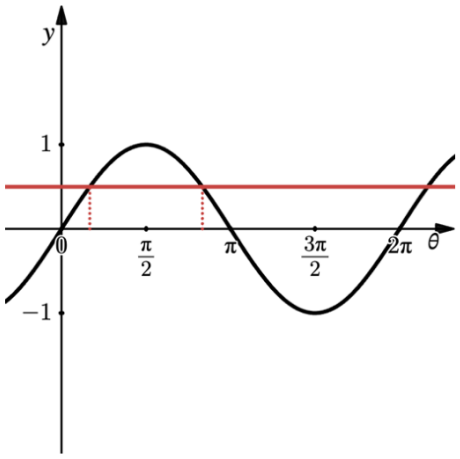
$$\theta = -0.7544\ldots - \pi = -3.8960\ldots \quad \text{(not in range)}$$
$$\theta = -3.8960\ldots - \pi = -7.0376\ldots \quad \text{(not in range)}$$
$$\theta = -7.0376\ldots - \pi = -10.1792\ldots \quad \text{(not in range)}$$
$$\theta = -10.1792\ldots - \pi = -13.3208\ldots \quad \text{(not in range)}$$
$$\theta = -13.3208\ldots - \pi = -16.4624\ldots \quad \text{(in range)}$$
$$\theta = -16.4624\ldots - \pi = -19.6040\ldots \quad \text{(in range)}$$
$$\Rightarrow \theta = \{-19.6, -16.5\} \text{ to 3 s.f.}$$

15 a)

$$\frac{1}{\sin\theta} = 16.5 \Rightarrow \sin\theta = \frac{1}{16.5} = \frac{2}{33} \text{ for } 0 \leq \theta < \pi.$$

Find first value between 0 and 2π:

$$\theta = \arcsin\frac{2}{33} = 0.0606\ldots \quad \text{(in range)}$$



Find second value between 0 and 2π:

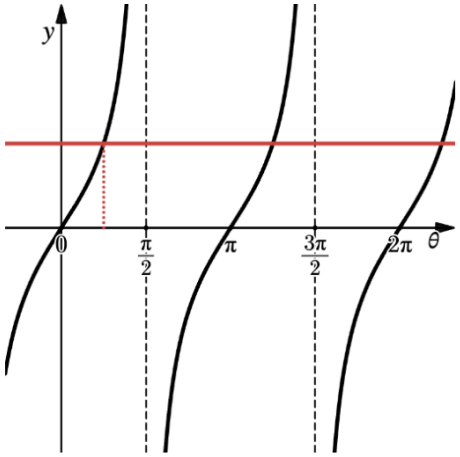
$$\theta = \pi - 0.0606\ldots = 3.0809\ldots \quad \text{(in range)}$$
$$\theta = \{0.0606, 3.08\} \text{ to 3 s.f.}$$

15 b)

$$\frac{14}{\tan\theta} = 19 \Rightarrow \tan\theta = \frac{14}{19} \text{ for } \pi \leq \theta < 4\pi.$$

Find first value:

$$\theta = \arctan\frac{14}{19} = 0.6350\ldots \quad \text{(not in range)}$$



Add period:

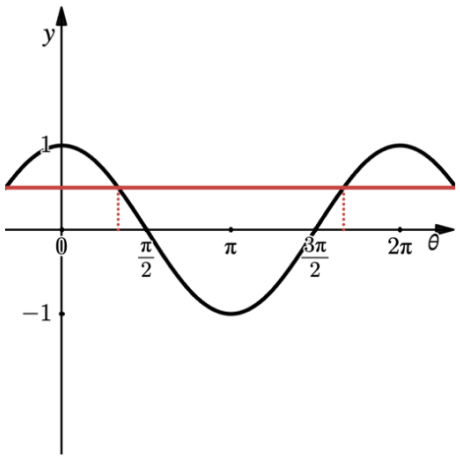
$$\theta = 0.6350\ldots + \pi = 3.7766\ldots \quad \text{(in range)}$$
$$\theta = 3.7766\ldots + \pi = 6.9182\ldots \quad \text{(in range)}$$
$$\theta = 6.9182\ldots + \pi = 10.0598\ldots \quad \text{(in range)}$$
$$\theta = \{3.78, 6.92, 10.1\} \text{ to 3 s.f.}$$

15 c)

$$\frac{0.02}{\cos\theta} = 0.3 \Rightarrow \cos\theta = \frac{0.02}{0.3} = \frac{1}{15} \text{ for } -\frac{5\pi}{2} \leq \theta < -\frac{3\pi}{2}.$$

Find first value between 0 and 2π:

$$\theta = \arccos\frac{1}{15} = 1.5040\ldots \quad \text{(in range)}$$



Find second value between 0 and 2π:

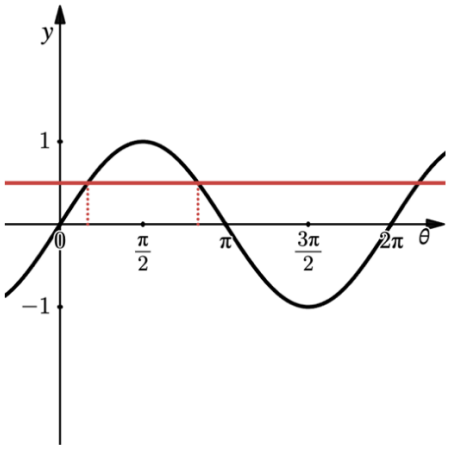
$$\theta = 2\pi - 1.5040\ldots = 4.7791\ldots \quad \text{(not in range)}$$

Subtract period:

$$\theta = 1.5040\ldots - 2\pi = -4.7791\ldots \quad \text{(in range)}$$
$$\theta = 4.7791\ldots - 2\pi = -1.5040\ldots \quad \text{(not in range)}$$
$$\theta = -1.5040\ldots - 2\pi = -7.7872\ldots \quad \text{(in range)}$$
$$\Rightarrow \theta = \{-7.79, -4.78\} \text{ to 3 s.f.}$$

16)

$$\frac{\sin A}{150} = \frac{\sin 1^{\text{c}}}{135} \Rightarrow A = \arcsin(\frac{150 \sin 1^{\text{c}}}{135}) = 1.2081\ldots \quad \text{(in range)}$$



Find second value between 0 and 2π:

$$A = \pi - 1.2081\ldots = 1.9334\ldots \quad \text{(in range)}$$
$$A = \{1.21, 1.93\} \text{ to 3 s.f.}$$

17)

$$A : x = -p$$
$$B : x = p - 2\pi$$
$$C : x = p - \pi$$
$$D : x = 2\pi - p.$$

18)

$$\sin \frac{\pi}{6} = \frac{1}{2}$$
$$\sin(\pi - \frac{\pi}{6}) = \sin \frac{5\pi}{6} = \frac{1}{2}$$
$$\sin(\frac{5\pi}{6} + 2\pi) = \sin \frac{17\pi}{6} = \frac{1}{2}.$$

19)

$$\cos \frac{\pi}{4} = \frac{\sqrt{2}}{2}$$
$$\cos(\frac{\pi}{2} + \frac{\pi}{4}) = \cos \frac{3\pi}{4} = -\frac{\sqrt{2}}{2}$$
$$\cos(\frac{3\pi}{4} - 8\pi) = \cos(-\frac{29\pi}{4}) = -\frac{\sqrt{2}}{2}.$$

20)

$$\tan(-\frac{39\pi}{4}) = 1$$
$$\tan(\frac{39\pi}{4}) = -1$$
$$\tan(\frac{39\pi}{4} - 5\pi) = \tan\frac{19\pi}{4} = -1.$$

21)

$$A : x = -p$$
$$B : x = p - 3\pi$$
$$C : x = \pi - p$$
$$D : x = p.$$

22)

$$A : x = p$$
$$B : x = 2\pi - p$$
$$C : x = \pi - p$$
$$D : x = -\pi - p.$$

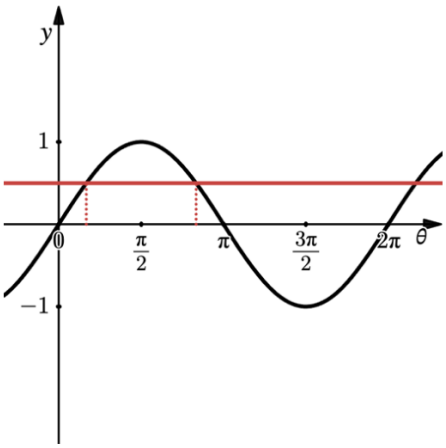
23 a)

$$\sin^2\theta = \frac{1}{25} \Rightarrow \sin\theta = \pm\frac{1}{5} \text{ for } 0 \leq \theta < 3\pi.$$

Solve $\sin\theta = \frac{1}{5}$ first:

Find first value between 0 and 2π:

$$\theta = \arcsin\frac{1}{5} = 0.2013\ldots \quad \text{(in range)}$$



Find second value between 0 and 2π:

$$\theta = \pi - 0.2013\ldots = 2.9402\ldots \quad \text{(in range)}$$

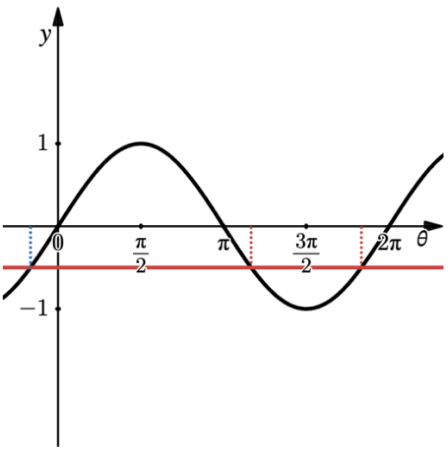
Add period:

$$\theta = 0.2013\ldots + 2\pi = 6.4845\ldots \quad \text{(in range)}$$
$$\theta = 2.9402\ldots + 2\pi = 9.2234\ldots \quad \text{(in range)}$$

Now solve $\sin\theta = -\frac{1}{5}$:

Find first value:

$$\theta = \arcsin(-\frac{1}{5}) = -0.2013\ldots \quad \text{(not in range)}$$



Find first value between 0 and 2π:

$$\theta = \pi + 0.2013\ldots = 3.3429\ldots \quad \text{(in range)}$$

Find second value between 0 and 2π:

$$\theta = 2\pi - 0.2013\ldots = 6.0818\ldots \quad \text{(in range)}$$
$$\theta = \{0.201, 2.94, 3.34, 6.08, 6.48, 9.22\} \text{ to 3 s.f.}$$

23 b)

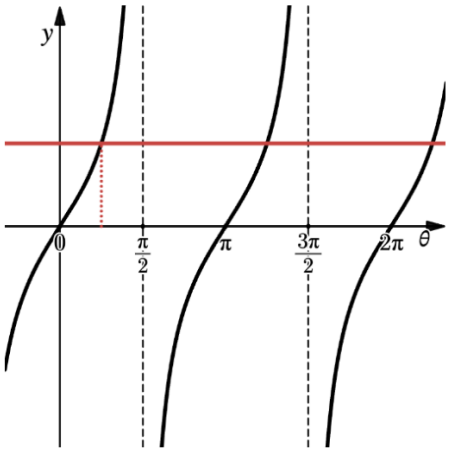
$$6\tan^2\theta = 19 \Rightarrow \tan\theta = \pm\sqrt{\frac{19}{6}} \text{ for } -3\pi \leq \theta < -\pi.$$

$-9.424\ldots < -3.141\ldots$.

Solve $\tan\theta = \sqrt{\frac{19}{16}}$ first:

Find first value:

$$\theta = \arctan\sqrt{\frac{19}{16}} = 0.8283\ldots \quad \text{(not in range)}$$



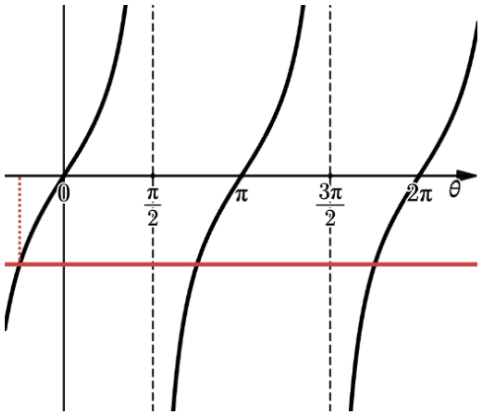
Subtract period:

$$\theta = 0.8283\ldots - \pi = -2.3132\ldots \quad \text{(not in range)}$$
$$\theta = -2.3132\ldots - \pi = -5.4548\ldots \quad \text{(in range)}$$
$$\theta = -5.4548\ldots - \pi = -8.5964\ldots \quad \text{(in range)}$$

Now solve $\tan\theta = -\sqrt{\frac{19}{16}}$:

Find first value:

$$\theta = \arctan(-\sqrt{\frac{19}{16}}) = -0.8283\ldots \quad \text{(not in range)}$$



Subtract period:

$$\theta = -0.8283\ldots - \pi = -3.9699\ldots \quad \text{(in range)}$$
$$\theta = -3.9699\ldots - \pi = -7.1114\ldots \quad \text{(in range)}$$
$$\theta = \{-8.60, -7.11, -5.45, -3.97\} \text{ to 3 s.f.}$$

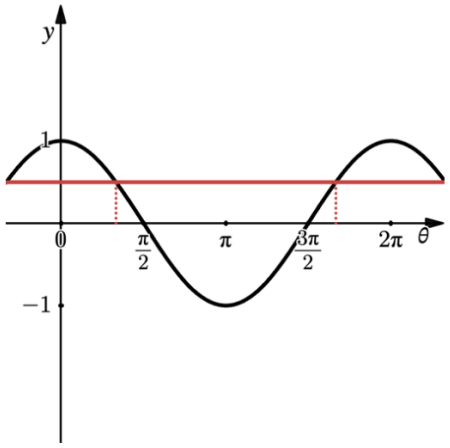
23 c)

$$25\cos^2\theta = \frac{1}{4} \Rightarrow \cos\theta = \pm\frac{1}{10} \text{ for } \pi \leq \theta < 4\pi.$$

Solve $\cos\theta = \frac{1}{10}$ first:

Find first value between 0 and 2π:

$$\theta = \arccos\frac{1}{10} = 1.4706\ldots \quad \text{(not in range)}$$



Find second value between 0 and 2π:

$$\theta = 2\pi - 1.4706\ldots = 4.8125\ldots \quad \text{(in range)}$$

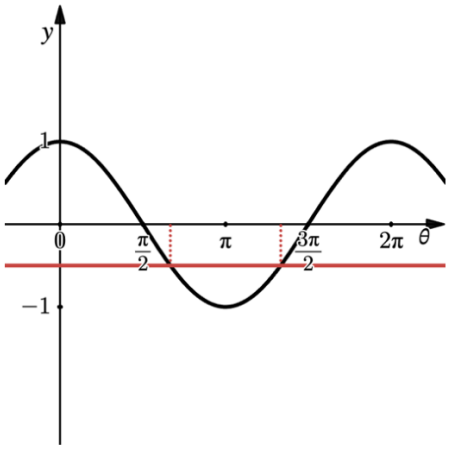
Add period:

$$\theta = 1.4706\ldots + 2\pi = 7.7538\ldots \quad \text{(in range)}$$
$$\theta = 4.8125\ldots + 2\pi = 11.0957\ldots \quad \text{(in range)}$$

Now solve $\cos\theta = -\frac{1}{10}$:

Find first value between 0 and 2π:

$$\theta = \arccos(-\frac{1}{10}) = 1.6709\ldots \quad \text{(not in range)}$$



Find second value between 0 and 2π:

$$\theta = 2\pi - 1.6709\ldots = 4.6122\ldots \quad \text{(in range)}$$

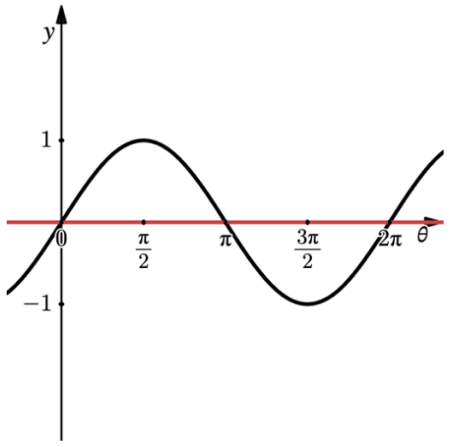
Add period:

$$\theta = 1.6709\ldots + 2\pi = 7.9541\ldots \quad \text{(in range)}$$
$$\theta = 4.6122\ldots + 2\pi = 10.8954\ldots \quad \text{(in range)}$$
$$\Rightarrow \theta = \{4.61, 4.81, 7.75, 7.95, 10.9, 11.1\} \text{ to 3 s.f.}$$

24 a)

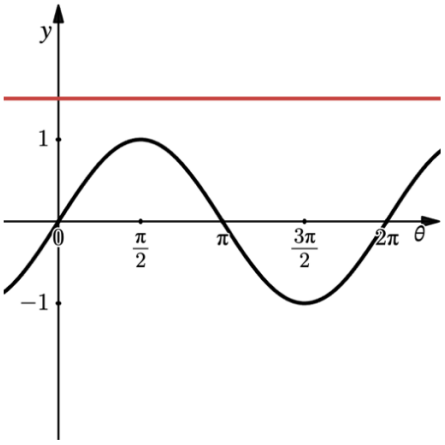
$$\sin^2\theta = 8\sin\theta \text{ for } 0 \leq \theta < 2\pi$$
$$\sin^2\theta - 8\sin\theta = 0 \Rightarrow \sin\theta(\sin\theta - 8) = 0$$
$$\Rightarrow \sin\theta = 0 \text{ or } \sin\theta = 8.$$

Solve $\sin\theta = 0$ first:



$$\theta = 0 \text{ or } \theta = \pi.$$

As for $\sin\theta = 8$, there are no solutions:

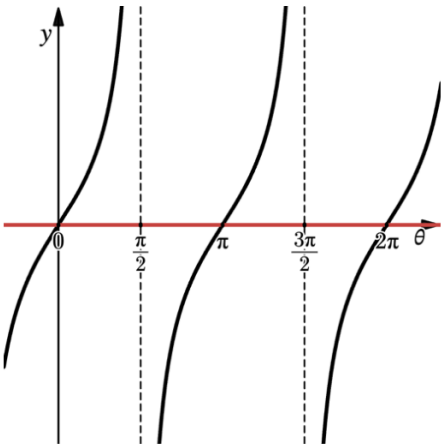


$$\theta = \{0, \pi\}.$$

24 b)

$$\begin{aligned} 9\tan^2\theta - 5\tan\theta = 0 \text{ for } 0 \leq \theta < 2\pi \\ \Rightarrow \tan\theta(9\tan\theta - 5) = 0 \\ \Rightarrow \tan\theta = 0 \text{ or } \tan\theta = \frac{5}{9}. \end{aligned}$$

Solve $\tan\theta = 0$ first:

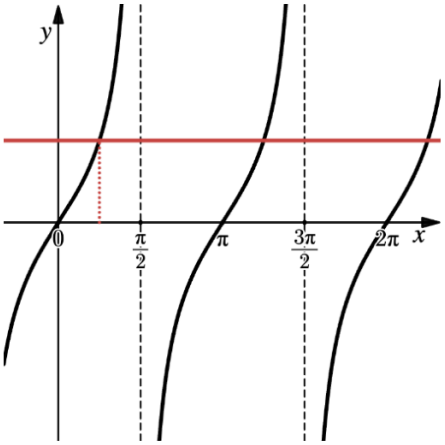


$$\theta = 0 \text{ or } \theta = \pi.$$

Now solve $\tan\theta = \frac{5}{9}$:

Find first value:

$$\theta = \arctan\frac{5}{9} = 0.5070\ldots \quad \text{(in range)}$$



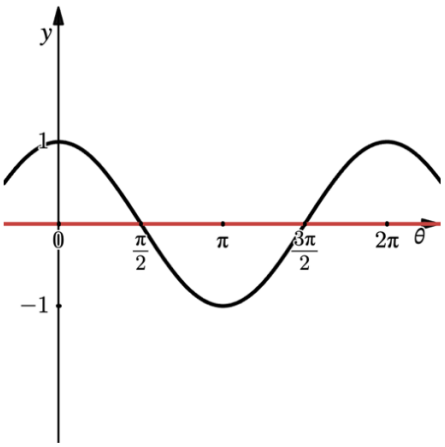
Add period:

$$\theta = 0.5070\ldots + \pi = 3.6486\ldots \quad \text{(in range)}$$
$$\theta = \{0, 0.507, \pi, 3.65\} \text{ to 3 s.f.}$$

24 c)

$$-6\cos\theta = 11\cos^2\theta \text{ for } 0 \leq \theta < 2\pi$$
$$\Rightarrow 11\cos^2\theta + 6\cos\theta = 0 \Rightarrow \cos\theta(11\cos\theta + 6) = 0$$
$$\Rightarrow \cos\theta = 0 \text{ or } \cos\theta = -\frac{6}{11}.$$

Solve $\cos\theta = 0$ first:

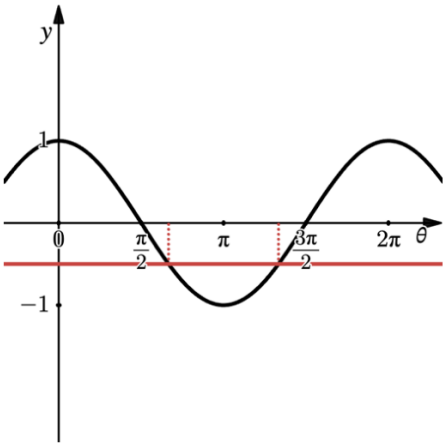


$$\theta = \frac{\pi}{2} \text{ or } \theta = \frac{3\pi}{2}.$$

Now solve $\cos\theta = -\frac{6}{11}$:

Find first value between 0 and 2π:

$$\theta = \arccos(-\frac{6}{11}) = 2.1477\ldots \quad \text{(in range)}$$



Find second value between 0 and 2π:

$$\theta = 2\pi - 2.1477\ldots = 4.1354\ldots \quad \text{(in range)}$$
$$\Rightarrow \theta = \{\frac{\pi}{2}, 2.15, 4.14, \frac{3\pi}{2}\} \text{ to 3 s.f.}$$

25)

$$A : x = p$$
$$B : x = p + \pi$$
$$C : x = p - \pi$$
$$D : x = p - 2\pi.$$

26)

$$A : x = -p$$
$$B : x = p - 3\pi$$
$$C : x = \pi - p$$
$$D : x = p.$$

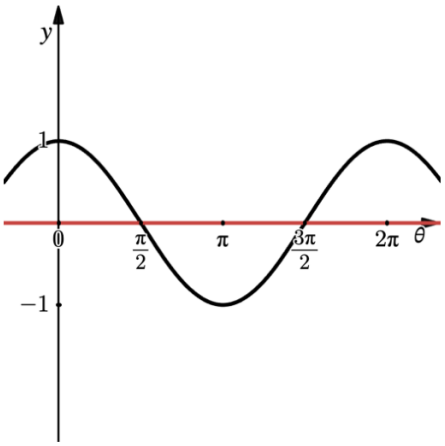
27)

$$A : x = -p$$
$$B : x = p - 2\pi$$
$$C : x = p - \pi$$
$$D : x = 3\pi - p.$$

28 a)

$$5\sin\theta\cos\theta = 3\cos\theta \text{ for } 0 \leq \theta < 2\pi$$
$$\Rightarrow 5\sin\theta\cos\theta - 3\cos\theta = 0$$
$$\Rightarrow \cos\theta(5\sin\theta - 3) = 0 \Rightarrow \cos\theta = 0 \text{ or } \sin\theta = \frac{3}{5}.$$

Solve $\cos\theta = 0$ first:

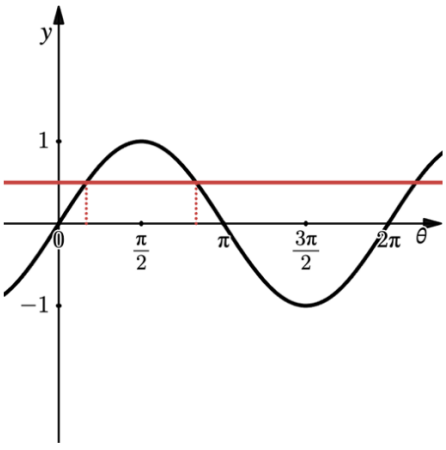


$$\theta = \frac{\pi}{2} \text{ or } \theta = \frac{3\pi}{2}.$$

Now solve $\sin\theta = \frac{3}{5}$:

Find first value between 0 and 2π:

$$\theta = \arcsin\frac{3}{5} = 0.6435\ldots \quad \text{(in range)}$$



Find second value between 0 and 2π:

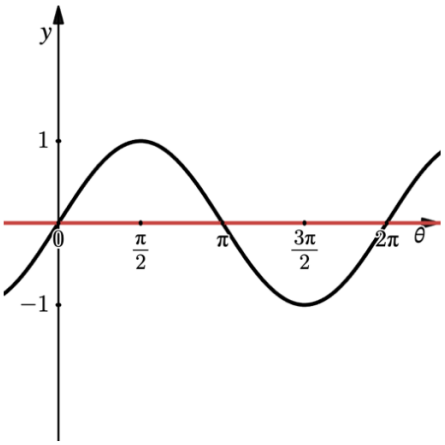
$$\theta = \pi - 0.6435\ldots = 2.4980\ldots \quad \text{(in range)}$$
$$\theta = \{0.644, \frac{\pi}{2}, 2.50, \frac{3\pi}{2}\} \text{ to 3 s.f.}$$

28 b)

$$\frac{2}{5}\sin\theta\tan\theta + 3\sin\theta = 0 \text{ for } 0 \leq \theta < 2\pi$$

$$\Rightarrow \sin\theta(\frac{2}{5}\tan\theta + 3) = 0 \Rightarrow \sin\theta = 0 \text{ or } \tan\theta = -\frac{15}{2}.$$

Solve $\sin\theta = 0$ first:

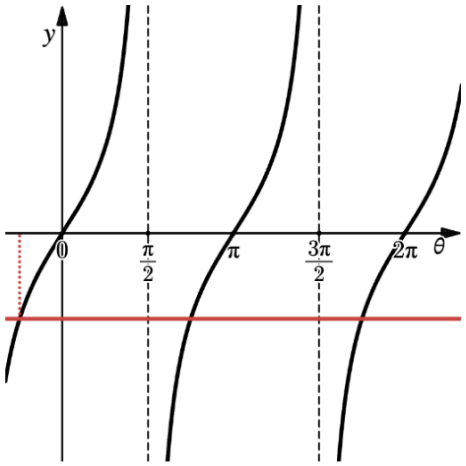


$$\theta = 0 \text{ or } \theta = \pi.$$

Now solve $\tan\theta = -\frac{15}{2}$:

Find first value:

$$\theta = \arctan(-\frac{15}{2}) = -1.4382\ldots \quad \text{(not in range)}$$



Add period:

$$\theta = -1.4382\ldots + \pi = 1.7033\ldots \quad \text{(in range)}$$

$$\theta = 1.7033\ldots + \pi = 4.8449\ldots \quad \text{(in range)}$$

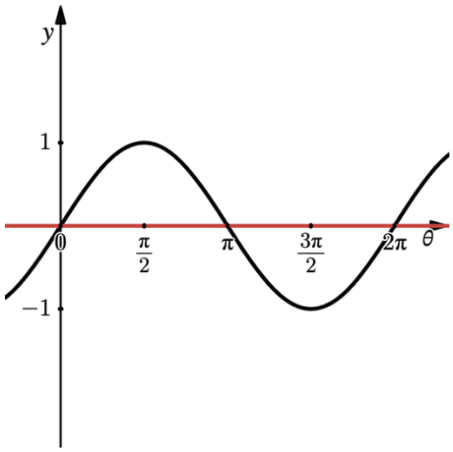
$$\theta = \{0, 1.70, \pi, 4.84\} \text{ to 3 s.f.}$$

28 c)

$$29\sin\theta - 31\sin\theta\cos\theta = 0 \text{ for } 0 \leq \theta < 2\pi$$

$$\Rightarrow \sin\theta(29 - 31\cos\theta) = 0 \Rightarrow \sin\theta = 0 \text{ or } \cos\theta = \frac{29}{31}.$$

Solve $\sin\theta = 0$ first:

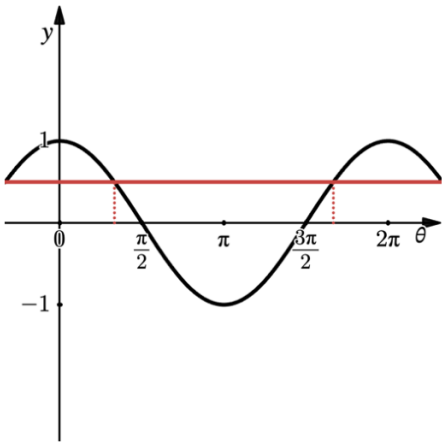


$$\theta = 0 \text{ or } \theta = \pi.$$

Now solve $\cos\theta = \frac{29}{31}$:

Find first value between 0 and 2π:

$$\theta = \arccos\frac{29}{31} = 0.3611\ldots \quad \text{(in range)}$$



Find second value between 0 and 2π:

$$\theta = 2\pi - 0.3611\ldots = 5.922\ldots \quad \text{(in range)}$$

$$\theta = \{0, 0.361, \pi, 5.92\} \text{ to 3 s.f.}$$

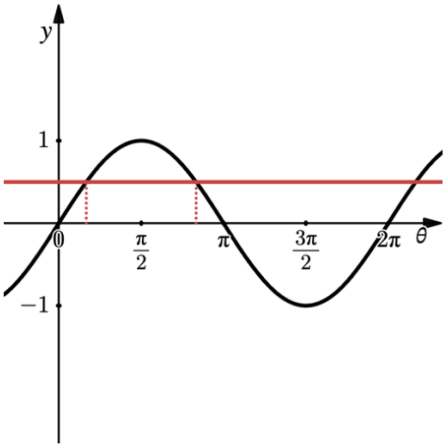
29 a)

$$6\sin^2\theta - \sin\theta - 1 = 0 \text{ for } 0 \le \theta < 2\pi$$
$$\Rightarrow (2\sin\theta - 1)(3\sin\theta + 1) = 0$$
$$\Rightarrow \sin\theta = \frac{1}{2} \text{ or } \sin\theta = -\frac{1}{3}.$$

Solve $\sin\theta = \frac{1}{2}$ first:

Find first value between 0 and 2π:

$$\theta = \arcsin\frac{1}{2} = \frac{\pi}{6} \quad \text{(in range)}$$



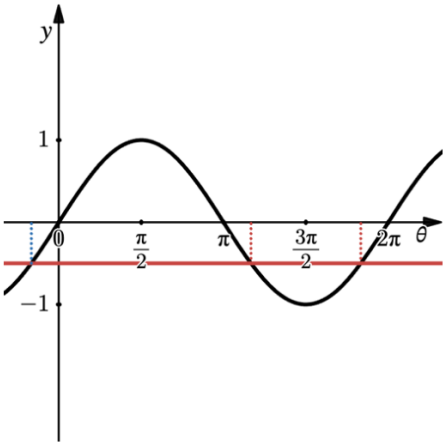
Find second value between 0 and 2π:

$$\theta = \pi - \frac{\pi}{6} = \frac{5\pi}{6} \quad \text{(in range)}$$

Now solve $\sin\theta = -\frac{1}{3}$:

Find first value:

$$\theta = \arcsin(-\frac{1}{3}) = -0.3398\ldots \quad \text{(not in range)}$$



Find first value between 0 and 2π:

$$\theta = \pi + 0.3398\ldots = 3.4814\ldots \quad \text{(in range)}$$

Find second value between 0 and 2π:

$$\theta = 2\pi - 0.3398\ldots = 5.9433\ldots \quad \text{(in range)}$$
$$\theta = \{\frac{\pi}{6}, \frac{5\pi}{6}, 3.48, 5.94\} \text{ to 3 s.f.}$$

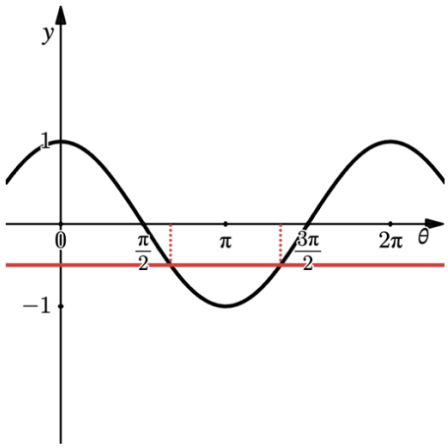
29 b)

$$15\cos^2\theta - 2\cos\theta - 1 = 0 \text{ for } 0 \leq \theta < 2\pi$$
$$\Rightarrow (5\cos\theta + 1)(3\cos\theta - 1) = 0$$
$$\Rightarrow \cos\theta = -\frac{1}{5} \text{ or } \cos\theta = \frac{1}{3}.$$

Solve $\cos\theta = -\frac{1}{5}$ first:

Find first value between 0 and 2π:

$$\theta = \arccos(-\frac{1}{5}) = 1.7721\ldots \quad \text{(in range)}$$



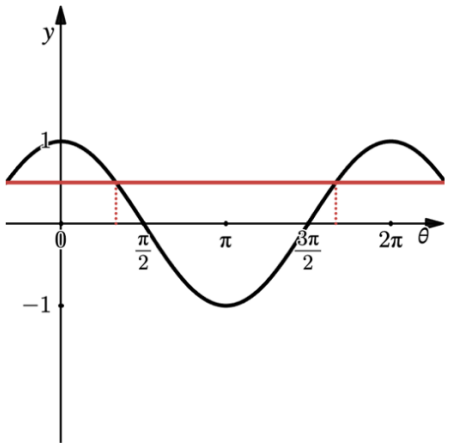
Find second value between 0 and 2π:

$$\theta = 2\pi - 1.7721\ldots = 4.5110\ldots \quad \text{(in range)}$$

Now solve $\cos\theta = \frac{1}{3}$:

Find first value between 0 and 2π:

$$\theta = \arccos\frac{1}{3} = 1.2309\ldots \quad \text{(in range)}$$



Find second value between 0 and 2π:

$$\theta = 2\pi - 1.2309\ldots = 5.0522\ldots \quad \text{(in range)}$$
$$\theta = \{1.23, 1.77, 4.51, 5.05\} \text{ to 3 s.f.}$$

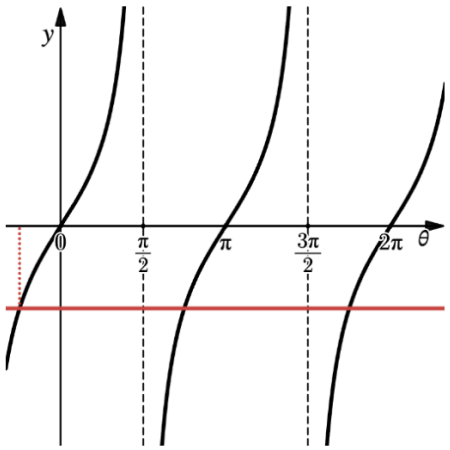
29 c)

$$18\tan^2\theta + 15\tan\theta + 2 = 0 \text{ for } 0 \leq \theta < 2\pi$$
$$\Rightarrow (6\tan\theta + 1)(3\tan\theta + 2) = 0$$
$$\Rightarrow \tan\theta = -\frac{1}{6} \text{ or } \tan\theta = -\frac{2}{3}.$$

Solve $\tan\theta = -\frac{1}{6}$ first:

Find first value:

$$\theta = \arctan(-\frac{1}{6}) = -0.1651\ldots \quad \text{(not in range)}$$



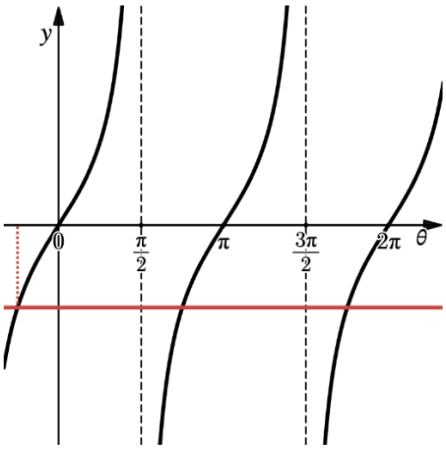
Add period:

$$\theta = -0.1651\ldots + \pi = 2.9764\ldots \quad \text{(in range)}$$
$$\theta = -2.9764\ldots + \pi = 6.1180\ldots \quad \text{(in range)}$$

Now solve $\tan\theta = -\frac{2}{3}$:

Find first value:

$$\theta = \arctan(-\frac{2}{3}) = -0.5880\ldots \quad \text{(not in range)}$$



Add period:

$$\theta = -0.5880\ldots + \pi = 2.5535\ldots \quad \text{(in range)}$$
$$\theta = 2.5535\ldots + \pi = 5.6951\ldots \quad \text{(in range)}$$
$$\theta = \{2.55, 2.98, 5.70, 6.12\} \text{ to 3 s.f.}$$

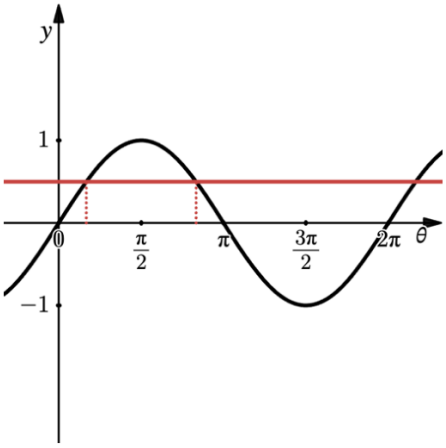
30 a)

$$\begin{aligned} 32\sin\theta = \frac{9}{\sin\theta} - 12 \text{ for } 0 \leq \theta < 2\pi \\ \Rightarrow 32\sin^2\theta = 9 - 12\sin\theta \\ \Rightarrow 32\sin^2\theta + 12\sin\theta - 9 = 0 \\ \Rightarrow (8\sin\theta - 3)(4\sin\theta + 3) = 0 \\ \Rightarrow \sin\theta = \frac{3}{8} \text{ or } \sin\theta = -\frac{3}{4}. \end{aligned}$$

Solve $\sin\theta = \frac{3}{8}$ first:

Find first value between 0 and 2π:

$$\theta = \arcsin\frac{3}{8} = 0.3843\ldots \quad \text{(in range)}$$



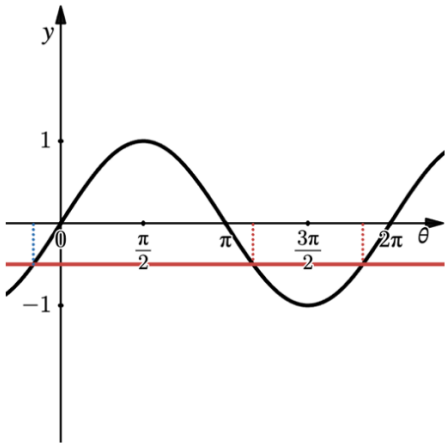
Find second value between 0 and 2π:

$$\theta = \pi - 0.3843\ldots = 2.7571\ldots \quad \text{(in range)}$$

Now solve $\sin\theta = -\frac{3}{4}$:

Find first value:

$$\theta = \arcsin(-\frac{3}{4}) = -0.8480\ldots \quad \text{(not in range)}$$



Find first value between 0 and 2π:

$$\theta = \pi + 0.8480\ldots = 3.9896\ldots \quad \text{(in range)}$$

Find second value between 0 and 2π:

$$\theta = 2\pi - 0.8480\ldots = 5.4351\ldots \quad \text{(in range)}$$

$$\theta = \{0.384, 2.76, 3.99, 5.44\} \text{ to 3 s.f.}$$

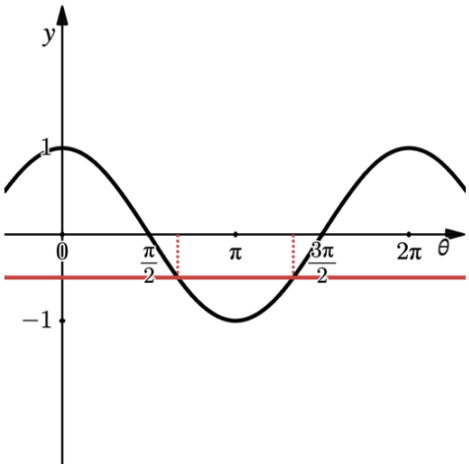
30 b)

$$\begin{aligned} 54\cos\theta = 51 + \frac{14}{\cos\theta} &\Rightarrow 54\cos^2\theta = 51\cos\theta + 14 \\ &\Rightarrow 54\cos^2\theta - 51\cos\theta - 14 = 0 \\ &\Rightarrow (9\cos\theta + 2)(6\cos\theta - 7) = 0 \\ &\Rightarrow \cos\theta = -\frac{2}{9} \text{ or } \cos\theta = \frac{7}{6}. \end{aligned}$$

Solve $\cos\theta = -\frac{2}{9}$ first:

Find first value between 0 and 2π:

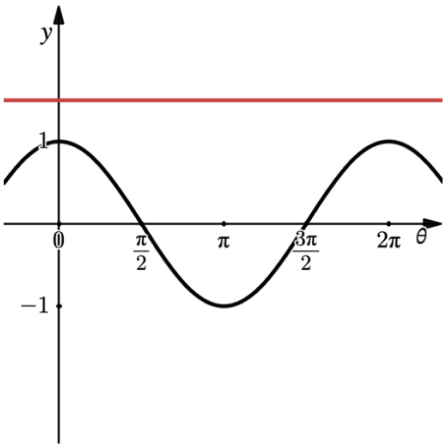
$$\theta = \arccos(-\frac{2}{9}) = 1.7948\ldots \quad \text{(in range)}$$



Find second value between 0 and 2π:

$$\theta = 2\pi - 1.7948\ldots = 4.4882\ldots \quad \text{(in range)}$$

As for $\cos\theta = \frac{7}{6}$, there are no solutions:



$$\theta = \{1.79, 4.49\} \text{ to 3 s.f.}$$

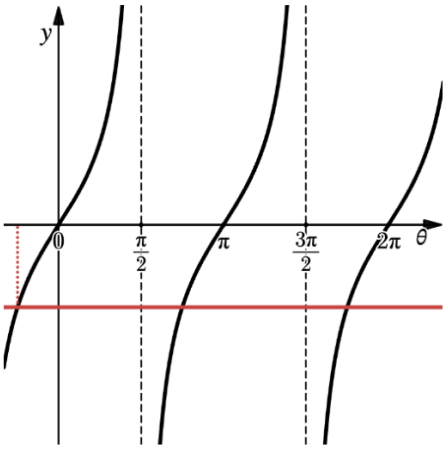
30 c)

$$20\tan\theta = 48 + \frac{77}{\tan\theta} \Rightarrow 20\tan^2\theta = 48\tan\theta + 77$$
$$\Rightarrow 20\tan^2\theta - 48\tan\theta - 77 = 0$$
$$\Rightarrow (10\tan\theta + 11)(2\tan\theta - 7) = 0$$
$$\Rightarrow \tan\theta = -\frac{11}{10} \text{ or } \tan\theta = \frac{7}{2}.$$

Solve $\tan\theta = -\frac{11}{10}$ first:

Find first value:

$$\theta = \arctan(-\frac{11}{10}) = -0.8329\ldots \quad \text{(not in range)}$$



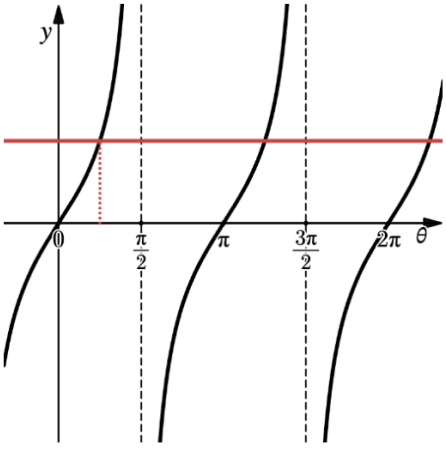
Add period:

$$\theta = -0.8329\ldots + \pi = 2.3086\ldots \quad \text{(in range)}$$
$$\theta = 2.3086\ldots + \pi = 5.4502\ldots \quad \text{(in range)}$$

Now solve $\tan\theta = \frac{7}{2}$:

Find first value:

$$\theta = \arctan\frac{7}{2} = 1.2924\ldots \quad \text{(in range)}$$



Add period:

$$\theta = 1.2924\ldots + \pi = 4.434\ldots \quad \text{(in range)}$$
$$\theta = \{1.29, 2.31, 4.43, 5.45\} \text{ to 3 s.f.}$$

31 a)

$$\begin{aligned} 5x^2 + 70x + 219 &= 5(x^2 + 14x) + 219 \\ &= 5((x+7)^2 - 49) + 219 \\ &= 5(x+7)^2 - 245 + 219 \\ &= 5(x+7)^2 - 26. \end{aligned}$$

31 b)

$$(-7, -26).$$

31 c)

$$5\cos^2\theta + 70\cos\theta + 219 = 5(\cos\theta + 7)^2 - 26.$$

The minimum value will occur when $(\cos\theta+7)^2$ is at its minimum. $-1 \leq \cos\theta \leq 1$, so the minimum will occur when $\cos\theta = -1$, which gives $5(-1+7)^2 - 26 = 154$. The smallest positive value of θ for which this minimum occurs is when $\theta = \pi$.

32 a)

Period is 2π.

32 b)

Period is 2π.

32 c)

Period is π.

32 d)

Period is $\frac{2\pi}{3}$.

32 e)

Period is $\frac{2\pi}{2} = \pi$.

32 f)

Period is $\frac{\pi}{6}$.

32 g)

Period is $\frac{2\pi}{1/5} = 10\pi$.

32 h)

Period is $\frac{2\pi}{8/3} = \frac{3\pi}{4}$.

32 i)

Period is $\frac{\pi}{9/16} = \frac{16\pi}{9}$.

33 a)

Period is $\frac{2\pi}{2} = \pi$ and expect two solutions each period:

$$\frac{2\pi - 0}{\pi} \times 2 = 4 \text{ solutions.}$$

33 b)

$$\frac{3\pi - 0}{\pi} \times 2 = 6 \text{ solutions.}$$

33 c)

$$\frac{8\pi - (-4\pi)}{\pi} \times 2 = 24 \text{ solutions.}$$

34 a)

Period is $\frac{2\pi}{2} = \pi$ and expect two solutions each period:

$$\frac{\pi - (-\pi)}{\pi} \times 2 = 4 \text{ solutions.}$$

34 b)

Period is $\frac{2\pi}{3}$ and expect two solutions each period:

$$\frac{5\pi - 0}{2\pi/3} \times 2 = 15 \text{ solutions.}$$

34 c)

Period is $\frac{2\pi}{4} = \frac{\pi}{2}$ and expect two solutions each period:

$$\frac{3\pi/2 - (-6\pi)}{\pi/2} \times 2 = 30 \text{ solutions.}$$

35 a)

Period is $\frac{\pi}{2}$ and expect **one** solution each period:

$$\frac{3\pi - (-2\pi)}{\pi/2} = 10 \text{ solutions.}$$

35 b)

Period is $\frac{\pi}{3}$ and expect **one** solution each period:

$$\frac{10\pi - (-5\pi)}{\pi/3} = 45 \text{ solutions.}$$

35 c)

Period is $\frac{\pi}{4}$ and expect **one** solution each period:

$$\frac{40\pi - (-25\pi)}{\pi/4} = 260 \text{ solutions.}$$

36 a)

Period is 2π.

36 b)

Period is 2π.

36 c)

Period is π.

36 d)

Period is 2π.

36 e)

Period is 2π.

36 f)

Period is π.

36 g)

Period is $\frac{2\pi}{3}$.

36 h)

Period is $\frac{2\pi}{6} = \frac{\pi}{3}$.

36 i)

Period is $\frac{\pi}{12}$.

36 j)

Period is $\frac{2\pi}{1/6} = 12\pi$.

36 k)

Period is $\frac{2\pi}{1/0.16} = \frac{8\pi}{25}$.

36 l)

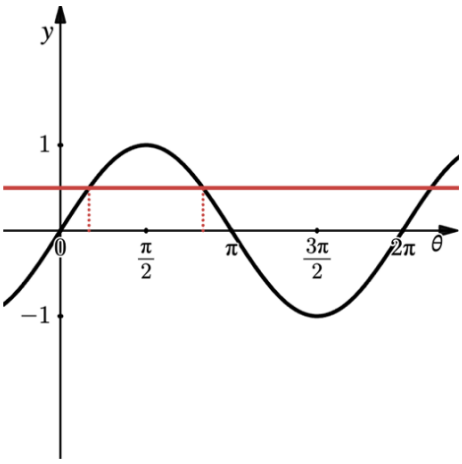
Period is $\frac{\pi}{15/12} = \frac{4\pi}{5}$.

37 a)

$\sin(\theta + 0.32) = 0.45$ for $0 \leq \theta < 2\pi$.

Find first value between 0 and 2π:

$$\theta + 0.32 = \arcsin 0.45 = 0.4667\ldots$$



Find second value between 0 and 2π:

$$\theta + 0.32 = \pi - 0.4667\ldots = 2.6748\ldots$$

Apply transformation:

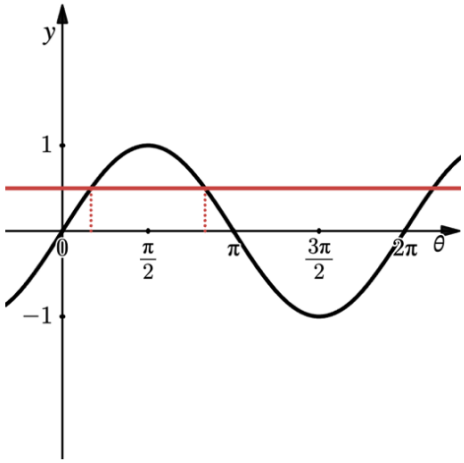
$$\begin{aligned}
\theta + 0.32 = 0.4667\ldots &\Rightarrow \theta = 0.1467\ldots \quad \text{(in range)} \\
\theta + 0.32 = 2.6748\ldots &\Rightarrow \theta = 2.3548\ldots \quad \text{(in range)} \\
&\Rightarrow \theta = \{0.147, 2.35\} \text{ to 3 s.f.}
\end{aligned}$$

37 b)

$\sin(\theta - \frac{8\pi}{9}) = \frac{6}{7}$ for $0 \leq \theta < 2\pi$.

Find first value between 0 and 2π:

$$\theta - \frac{8\pi}{9} = \arcsin \frac{6}{7} = 1.0296\ldots$$



Find second value between 0 and 2π:

$$\theta - \frac{8\pi}{9} = \pi - 1.0296\ldots = 2.1118\ldots$$

Apply transformation:

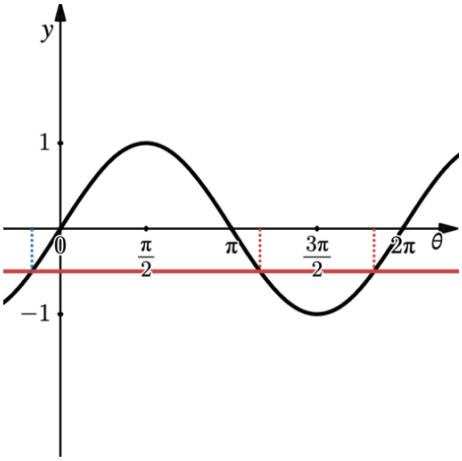
$$\theta - \frac{8\pi}{9} = 1.0296\ldots \Rightarrow \theta = 3.8222\ldots \quad \text{(in range)}$$
$$\theta - \frac{8\pi}{9} = 2.1118\ldots \Rightarrow \theta = 4.9044\ldots \quad \text{(in range)}$$
$$\theta = \{3.82, 4.90\} \text{ to 3 s.f.}$$

37 c)

$\sin(\theta - 1.2) = -0.33$ for $0 \leq \theta < 2\pi$.

Find first value:

$$\theta - 1.2 = \arcsin(-0.33) = -0.3363\ldots$$



Find first value between 0 and 2π:

$$\theta - 1.2 = \pi + 0.3363\ldots = 3.4778\ldots$$

Find second value between 0 and 2π:

$$\theta - 1.2 = 2\pi - 0.3363\ldots = 5.9468\ldots$$

Apply transformation:

$$\theta - 1.2 = 3.4778\ldots \Rightarrow \theta = 4.6778\ldots \quad \text{(in range)}$$
$$\theta - 1.2 = 5.9468\ldots \Rightarrow \theta = 7.1468\ldots \quad \text{(not in range)}$$

Subtract period:

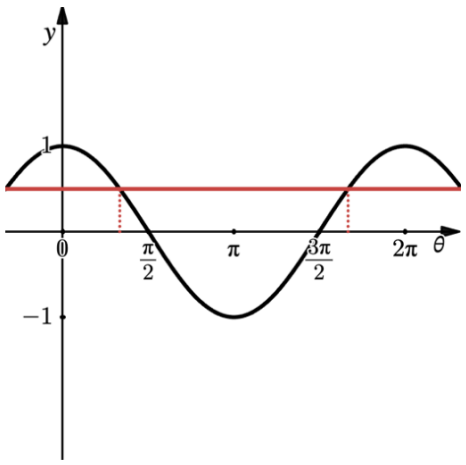
$$\theta = 7.1468\ldots - 2\pi = 0.8636\ldots \quad \text{(in range)}$$
$$\theta = \{0.864, 4.68\} \text{ to 3 s.f.}$$

38 a)

$\cos(\theta - 0.97) = 0.2$ for $0 \leq \theta < 2\pi$.

Find first value between 0 and 2π:

$$\theta - 0.97 = \arccos 0.2 = 1.3694\ldots$$



Find second value between 0 and 2π:

$$\theta - 0.97 = 2\pi - 1.3694\ldots = 4.9137\ldots$$

Apply transformation:

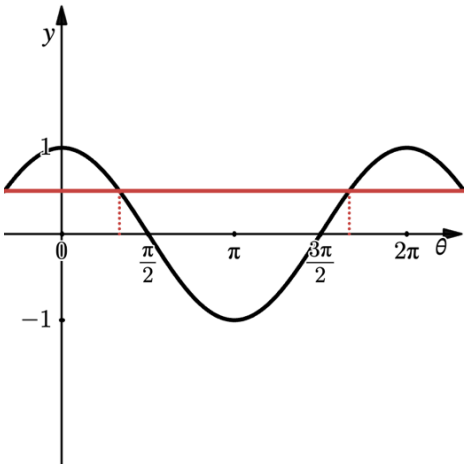
$$\begin{aligned}
\theta - 0.97 = 1.3694\ldots &\Rightarrow \theta = 2.3394\ldots \quad \text{(in range)} \\
\theta - 0.97 = 4.9137\ldots &\Rightarrow \theta = 5.8837\ldots \quad \text{(in range)} \\
\theta &= \{2.34, 5.88\} \text{ to 3 s.f.}
\end{aligned}$$

38 b)

$\cos(\theta + 2.3) = 0.49$ for $0 \leq \theta < 2\pi$.

Find first value between 0 and 2π:

$$\theta + 2.3 = \arccos 0.49 = 1.0587\ldots$$



Find second value between 0 and 2π:

$$\theta + 2.3 = 2\pi - 1.0587\ldots = 5.2244\ldots$$

Apply transformation:

$$\theta + 2.3 = 1.0587\ldots \Rightarrow \theta = -1.2412\ldots \quad \text{(not in range)}$$

$$\theta + 2.3 = 5.2244\ldots \Rightarrow \theta = 2.9244\ldots \quad \text{(in range)}$$

Add period:

$$\theta = -1.2412\ldots + 2\pi = 5.0418\ldots \quad \text{(in range)}$$

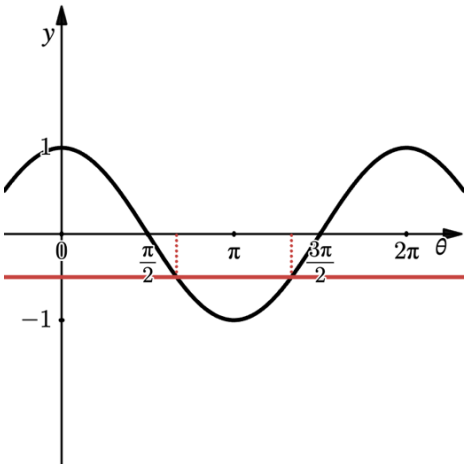
$\theta = \{2.92, 5.04\}$ to 3sf.

38 c)

$\cos(\theta - \frac{5\pi}{3}) = -0.89$ for $0 \leq \theta < 2\pi$.

Find first value between 0 and 2π:

$$\theta - \frac{5\pi}{3} = \arccos(-0.89) = 2.6681\ldots$$



Find second value between 0 and 2π:

$$\theta - \frac{5\pi}{3} = 2\pi - 2.6681\ldots = 3.6150\ldots$$

Apply transformation:

$$\theta - \frac{5\pi}{3} = 2.6681\ldots \Rightarrow \theta = 7.9041\ldots \quad \text{(not in range)}$$

$$\theta - \frac{5\pi}{3} = 3.6150\ldots \Rightarrow \theta = 8.8510\ldots \quad \text{(not in range)}$$

Subtract period:

$$\theta = 7.9041\ldots - 2\pi = 1.6209\ldots \quad \text{(in range)}$$

$$\theta = 8.8510\ldots - 2\pi = 2.5678\ldots \quad \text{(in range)}$$

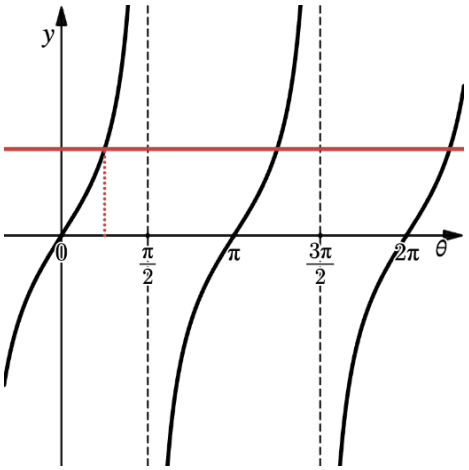
$\theta = \{1.62, 2.57\}$ to 3sf.

39 a)

$\tan(\theta + 2.01) = 6$ for $0 \leq \theta < 2\pi$.

Find first value:

$$\theta + 2.01 = \arctan 6 = 1.4056\ldots$$



Apply transformation:

$$\theta + 2.01 = 1.4056\ldots \Rightarrow \theta = -0.6043\ldots \quad \text{(not in range)}$$

Add period:

$$\theta = -0.6043\ldots + \pi = 2.5372\ldots \quad \text{(in range)}$$

$$\theta = 2.5372\ldots + \pi = 5.6788\ldots \quad \text{(in range)}$$

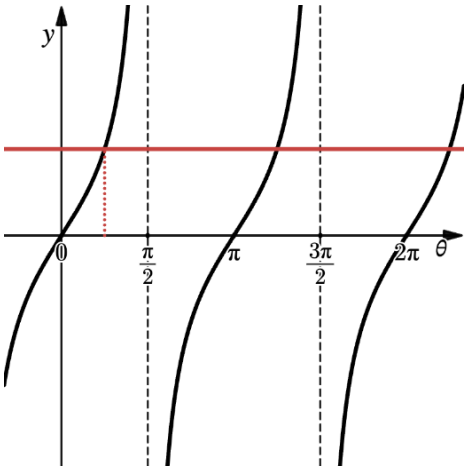
$\theta = \{2.54, 5.68\}$ to 3sf.

39 b)

$\tan(\theta - \frac{\pi}{5}) = \frac{3}{8}$ for $0 \leq \theta < 2\pi$.

Find first value:

$$\theta - \frac{\pi}{5} = \arctan\frac{3}{8} = 0.3587\ldots$$



Apply transformation:

$$\theta - \frac{\pi}{5} = 0.3587\ldots \Rightarrow \theta = 0.9870\ldots^\circ \quad \text{(in range)}$$

Add period:

$$\theta = 0.9870\ldots + \pi = 4.1286\ldots \quad \text{(in range)}$$

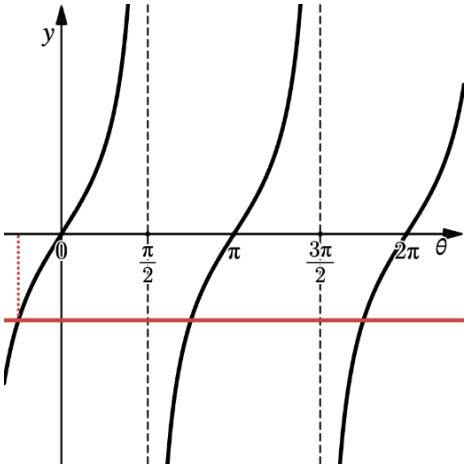
$\theta = \{0.987, 4.13\}$ to 3sf.

39 c)

$\tan(\theta + 0.99) = -8.2$ for $0 \leq \theta < 2\pi$.

Find first value:

$$\theta + 0.99 = \arctan(-8.2) = -1.4494\ldots$$



Apply transformation:

$$\theta + 0.99 = -1.4494\ldots \Rightarrow \theta = -2.4394\ldots \quad \text{(not in range)}$$

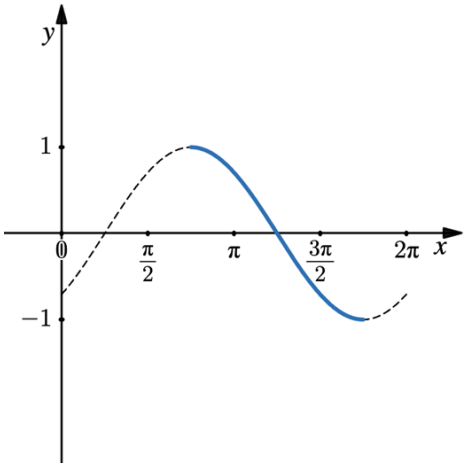
Add period:

$$\theta = -2.4394\ldots + \pi = 0.7021\ldots \quad \text{(in range)}$$

$$\theta = 0.7021\ldots + \pi = 3.8437\ldots \quad \text{(in range)}$$

$\theta = \{0.702, 3.84\}$ to 3sf.

40)



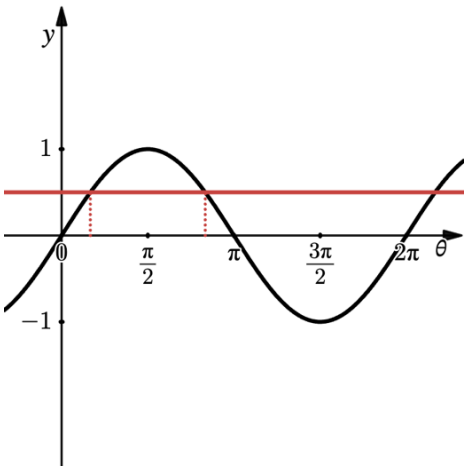
$\frac{3\pi}{4} < x < \frac{7\pi}{4}$.

41 a)

$\sin 2\theta = \frac{\sqrt{2}}{2}$ for $0 \leq \theta < 2\pi$.

Find first value between 0 and 2π:

$$2\theta = \arcsin \frac{\sqrt{2}}{2} = \frac{\pi}{4}$$



Find second value between 0 and 2π:

$$2\theta = \pi - \frac{\pi}{4} = \frac{3\pi}{4}$$

Apply transformation:

$$2\theta = \frac{\pi}{4} \Rightarrow \theta = \frac{\pi}{8} \quad \text{(in range)}$$

$$2\theta = \frac{3\pi}{4} \Rightarrow \theta = \frac{3\pi}{8} \quad \text{(in range)}$$

Add period:

$$\theta = \frac{\pi}{8} + \pi = \frac{9\pi}{8} \quad \text{(in range)}$$

$$\theta = \frac{3\pi}{8} + \pi = \frac{11\pi}{8} \quad \text{(in range)}$$

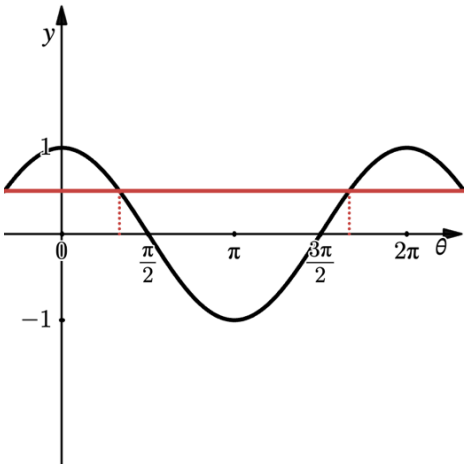
$\theta = \{\frac{\pi}{8}, \frac{3\pi}{8}, \frac{9\pi}{8}, \frac{11\pi}{8}\}$.

41 b)

$\cos 2\theta = \frac{\sqrt{3}}{2}$ for $0 \leq \theta < 2\pi$.

Find first value between 0 and 2π:

$$2\theta = \arccos \frac{\sqrt{3}}{2} = \frac{\pi}{6}$$



Find second value between 0 and 2π:

$$2\theta = 2\pi - \frac{\pi}{6} = \frac{11\pi}{6}$$

Apply transformation:

$$2\theta = \frac{\pi}{6} \Rightarrow \theta = \frac{\pi}{12} \quad \text{(in range)}$$

$$2\theta = \frac{11\pi}{6} \Rightarrow \theta = \frac{11\pi}{12} \quad \text{(in range)}$$

Add period:

$$\frac{\pi}{12} + \pi = \frac{13\pi}{12} \quad \text{(in range)}$$

$$\frac{11\pi}{12} + \pi = \frac{23\pi}{12} \quad \text{(in range)}$$

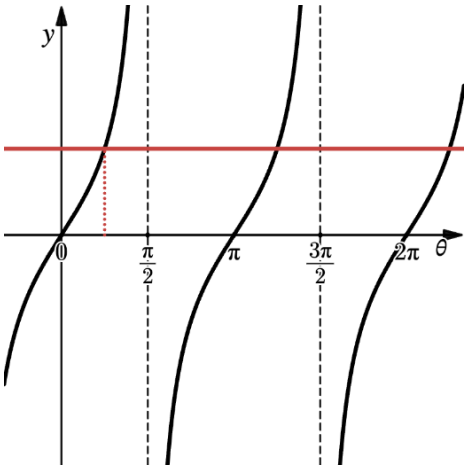
$\theta = \{\frac{\pi}{12}, \frac{11\pi}{12}, \frac{13\pi}{12}, \frac{23\pi}{12}\}$.

41 c)

$\tan 2\theta = \sqrt{3}$ for $0 \leq \theta < 2\pi$.

Find first value:

$$2\theta = \arctan \sqrt{3} = \frac{\pi}{3}$$



Apply transformation:

$$2\theta = \frac{\pi}{3} \Rightarrow \theta = \frac{\pi}{6} \quad \text{(in range)}$$

Add period:

$$\theta = \frac{\pi}{6} + \frac{\pi}{2} = \frac{2\pi}{3} \quad \text{(in range)}$$

$$\theta = \frac{2\pi}{3} + \frac{\pi}{2} = \frac{7\pi}{6} \quad \text{(in range)}$$

$$\theta = \frac{7\pi}{6} + \frac{\pi}{2} = \frac{5\pi}{3} \quad \text{(in range)}$$

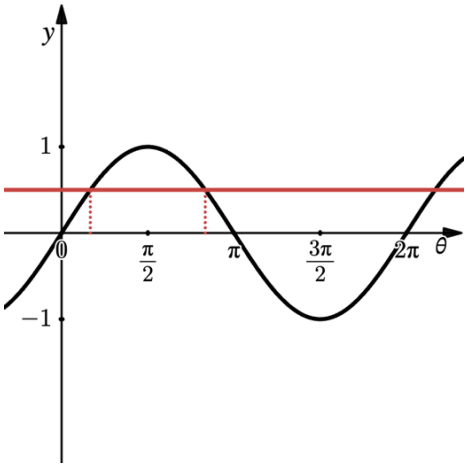
$\theta = \{\frac{\pi}{6}, \frac{2\pi}{3}, \frac{7\pi}{6}, \frac{5\pi}{3}\}$.

42 a)

$\sin 3\theta = 0.76$ for $0 \leq \theta < 2\pi$.

Find first value between 0 and 2π:

$$3\theta = \arcsin 0.76 = 0.8633\ldots$$



Find second value between 0 and 2π:

$$3\theta = \pi - 0.8633\ldots = 2.2782\ldots$$

Apply transformation:

$$3\theta = 0.8633\ldots \Rightarrow \theta = 0.2877\ldots \quad \text{(in range)}$$

$$3\theta = 2.2782\ldots \Rightarrow \theta = 0.7594\ldots \quad \text{(in range)}$$

Add period:

$$\theta = 0.2877\ldots + \frac{2\pi}{3} = 2.382\ldots \quad \text{(in range)}$$

$$\theta = 0.7594\ldots + \frac{2\pi}{3} = 2.8538\ldots \quad \text{(in range)}$$

$$\theta = 2.382\ldots + \frac{2\pi}{3} = 4.4765\ldots \quad \text{(in range)}$$

$$\theta = 2.8538\ldots + \frac{2\pi}{3} = 4.9482\ldots \quad \text{(in range)}$$

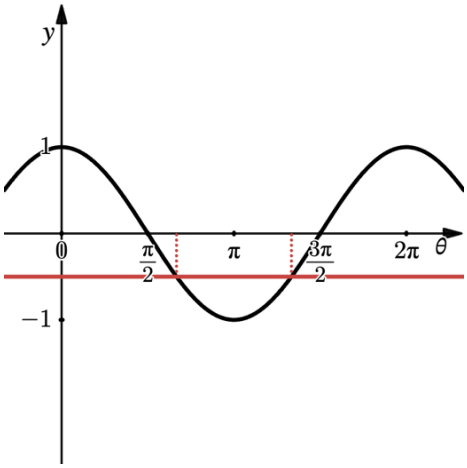
$\theta = \{0.288, 0.759, 2.38, 2.85, 4.48, 4.95\}$ to 3sf.

42 b)

$\cos 3\theta = -0.95$ for $0 \leq \theta < 2\pi$.

Find first value between 0 and 2π:

$$3\theta = \arccos(-0.95) = 2.8240\ldots$$



Find second value between 0 and 2π:

$$3\theta = 2\pi - 2.8240\ldots = 3.4591\ldots$$

Apply transformation:

$$3\theta = 2.8240\ldots \Rightarrow \theta = 0.9413\ldots \quad \text{(in range)}$$

$$3\theta = 3.4591\ldots \Rightarrow \theta = 1.1530\ldots \quad \text{(in range)}$$

Add period:

$$\theta = 0.9413\ldots + \frac{2\pi}{3} = 3.0357\ldots \quad \text{(in range)}$$

$$\theta = 1.1530\ldots + \frac{2\pi}{3} = 3.2474\ldots \quad \text{(in range)}$$

$$\theta = 3.0357\ldots + \frac{2\pi}{3} = 5.1301\ldots \quad \text{(in range)}$$

$$\theta = 3.2474\ldots + \frac{2\pi}{3} = 5.3418\ldots \quad \text{(in range)}$$

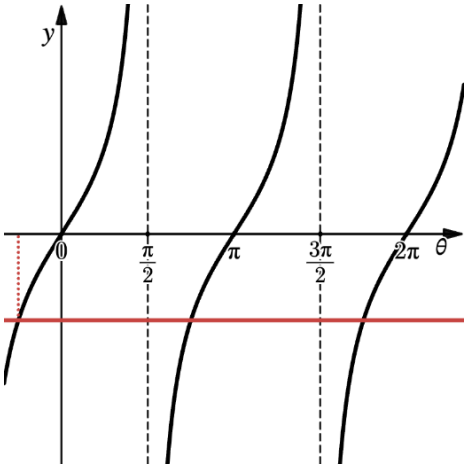
$\theta = \{0.941, 1.15, 3.04, 3.25, 5.13, 5.34\}$ to 3sf.

42 c)

$\tan 3\theta = -0.32$ for $0 \leq \theta < 2\pi$.

Find first value:

$$3\theta = \arctan(-0.32) = -0.3097\ldots$$



Apply transformation:

$$3\theta = -0.3097\ldots \Rightarrow \theta = -0.1032\ldots \quad \text{(not in range)}$$

Add period:

$$\theta = -0.1032\ldots + \frac{\pi}{3} = 0.9439\ldots \quad \text{(in range)}$$

$$\theta = 0.9439\ldots + \frac{\pi}{3} = 1.9911\ldots \quad \text{(in range)}$$

$$\theta = 1.9911\ldots + \frac{\pi}{3} = 3.0383\ldots \quad \text{(in range)}$$

$$\theta = 3.0383\ldots + \frac{\pi}{3} = 4.0855\ldots \quad \text{(in range)}$$

$$\theta = 4.0855\ldots + \frac{\pi}{3} = 5.1327\ldots \quad \text{(in range)}$$

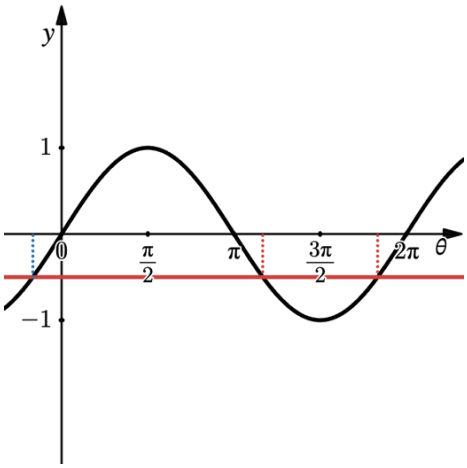
$$\theta = 5.1327\ldots + \frac{\pi}{3} = 6.1799\ldots \quad \text{(in range)}$$

$\theta = \{0.944, 1.99, 3.04, 4.09, 5.13, 6.18\}$ to 3sf.

43 a)

$\sin \frac{1}{2}\theta = -0.28$ for $0 \leq \theta < 4\pi$.

Find first value: $\frac{1}{2}\theta = \arcsin(-0.28) = -0.2837\ldots$.



Find first value between 0 and 2π: $\frac{1}{2}\theta = \pi + 0.2837\ldots = 3.4253\ldots$.

Find second value between 0 and 2π: $\frac{1}{2}\theta = 2\pi - 0.2837\ldots = 5.9993\ldots$.

Apply transformation:

$$\tfrac{1}{2}\theta = 3.4253\ldots \Rightarrow \theta = 6.8507\ldots \quad \text{(in range)}$$

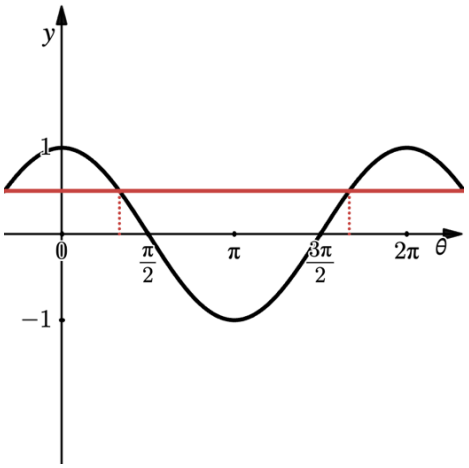
$$\tfrac{1}{2}\theta = 5.9993\ldots \Rightarrow \theta = 11.9987\ldots \quad \text{(in range)}$$

$\theta = \{6.85, 12.0\}$ to 3sf.

43 b)

$\cos \frac{1}{5}\theta = \frac{2}{9}$ for $0 \leq \theta < 10\pi$.

Find first value between 0 and 2π: $\frac{1}{5}\theta = \arccos \frac{2}{9} = 1.3467\ldots$.



Find second value between 0 and 2π: $\frac{1}{5}\theta = 2\pi - 1.3467\ldots = 4.9364\ldots$.

Apply transformation:

$$\frac{1}{5}\theta = 1.3467\ldots \Rightarrow \theta = 6.7335\ldots \quad \text{(in range)}$$

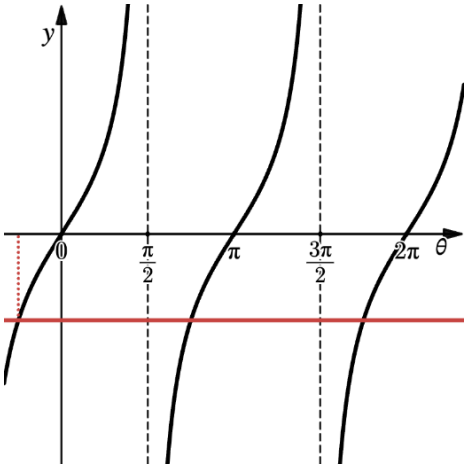
$$\frac{1}{5}\theta = 4.9364\ldots \Rightarrow \theta = 24.6824\ldots \quad \text{(in range)}$$

$\theta = \{6.73, 24.7\}$ to 3sf.

43 c)

$\tan \frac{7}{5}\theta = -2.4$ for $0 \leq \theta < \pi$.

Find first value: $\frac{7}{5}\theta = \arctan(-2.4) = -1.1760\ldots$.



Apply transformation: $\frac{7}{5}\theta = -1.1760\ldots \Rightarrow \theta = -0.8400\ldots$ (not in range).

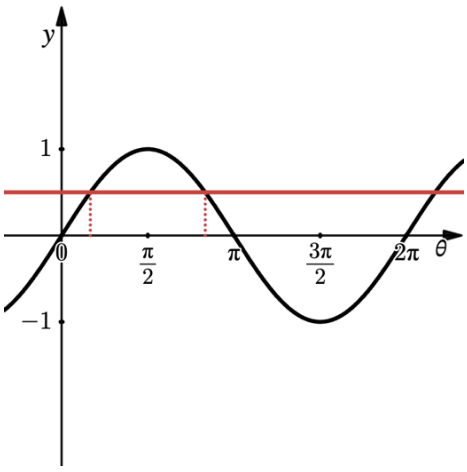
Add period: $\theta = -0.8400\ldots + \frac{5\pi}{7} = 1.4039\ldots$ (in range).

$\theta = \{1.40\}$ to 3sf.

44 a)

$\sin(3\theta - 0.31) = \frac{2}{7}$ for $0 \leq \theta < \pi$.

Find first value between 0 and 2π: $3\theta - 0.31 = \arcsin \frac{2}{7} = 0.2897\ldots$.



Find second value between 0 and 2π: $3\theta - 0.31 = \pi - 0.2897\ldots = 2.8518\ldots$.

Apply transformation:

$$3\theta - 0.31 = 0.2897\ldots \Rightarrow \theta = 0.1999\ldots \quad \text{(in range)}$$

$$3\theta - 0.31 = 2.8518\ldots \Rightarrow \theta = 1.0539\ldots \quad \text{(in range)}$$

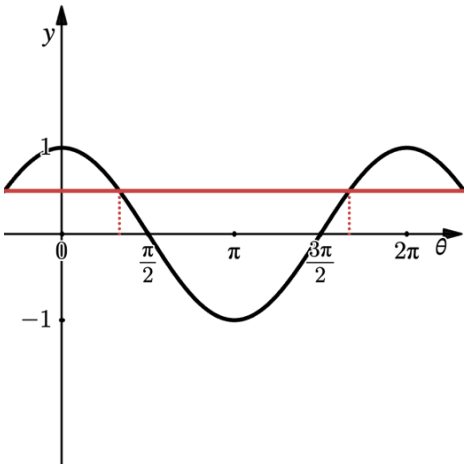
Add period: $\theta = 0.1999\ldots + \frac{2\pi}{3} = 2.2943\ldots$ (in range).

$\theta = \{0.200, 1.05, 2.29\}$ to 3sf.

44 b)

$\cos(4\theta + 0.97) = \frac{6}{11}$ for $0 \leq \theta < \pi$.

Find first value between 0 and 2π: $4\theta + 0.97 = \arccos\frac{6}{11} = 0.9938\ldots$.



Find second value between 0 and 2π: $4\theta + 0.97 = 2\pi - 0.9938\ldots = 5.2893\ldots$.

Apply transformation:

$$4\theta + 0.97 = 0.9938\ldots \Rightarrow \theta = 0.0059\ldots \quad \text{(in range)}$$

$$4\theta + 0.97 = 5.2893\ldots \Rightarrow \theta = 1.0798\ldots \quad \text{(in range)}$$

Add period:

$$\theta = 0.0059\ldots + \frac{\pi}{2} = 1.5767\ldots \quad \text{(in range)}$$

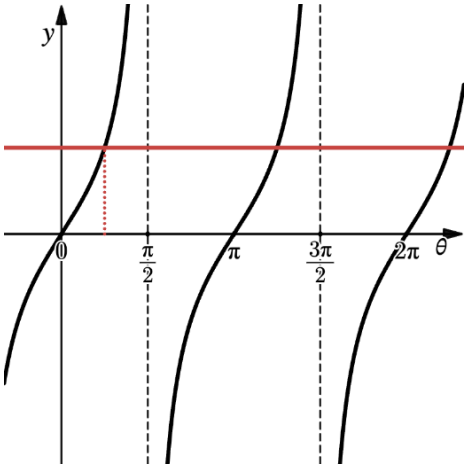
$$\theta = 1.0798\ldots + \frac{\pi}{2} = 2.6506\ldots \quad \text{(in range)}$$

$\theta = \{0.00597, 1.080, 1.58, 2.65\}$ to 3sf.

44 c)

$\tan(5\theta + 1.45) = \frac{7}{8}$ for $0 \leq \theta < \frac{\pi}{2}$.

Find first value: $5\theta + 1.45 = \arctan\frac{7}{8} = 0.7188\ldots$.



Apply transformation: $5\theta + 1.45 = 0.7188\ldots \Rightarrow \theta = -0.1462\ldots$ (not in range).

Add period:

$$\theta = -0.1462\ldots + \frac{\pi}{5} = 0.4820\ldots \quad \text{(in range)}$$

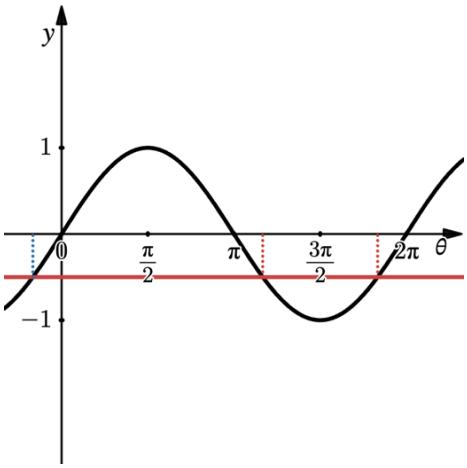
$$\theta = 0.4820\ldots + \frac{\pi}{5} = 1.1104\ldots \quad \text{(in range)}$$

$\theta = \{0.482, 1.11\}$ to 3sf.

45 a)

$\sin(5\theta + 3.2) = -0.16$ for $-\pi \leq \theta < 0$.

Find first value: $5\theta + 3.2 = \arcsin(-0.16) = -0.1606\ldots$.



Find first value between 0 and 2π: $5\theta + 3.2 = \pi + 0.1606\ldots = 3.3022\ldots$.

Find second value between 0 and 2π: $5\theta + 3.2 = 2\pi - 0.1606\ldots = 6.1224\ldots$.

Apply transformation:

$$5\theta + 3.2 = 3.3022\ldots \Rightarrow \theta = 0.0204\ldots \quad \text{(not in range)}$$

$$5\theta + 3.2 = 6.1224\ldots \Rightarrow \theta = 0.5844\ldots \quad \text{(not in range)}$$

Subtract period:

$$\theta = 0.0204\ldots - \frac{2\pi}{5} = -1.2361\ldots \quad \text{(in range)}$$

$$\theta = -1.2361\ldots - \frac{2\pi}{5} = -2.4928\ldots \quad \text{(in range)}$$

$$\theta = 0.5844\ldots - \frac{2\pi}{5} = -0.6721\ldots \quad \text{(in range)}$$

$$\theta = -0.6721\ldots - \frac{2\pi}{5} = -1.9287\ldots \quad \text{(in range)}$$

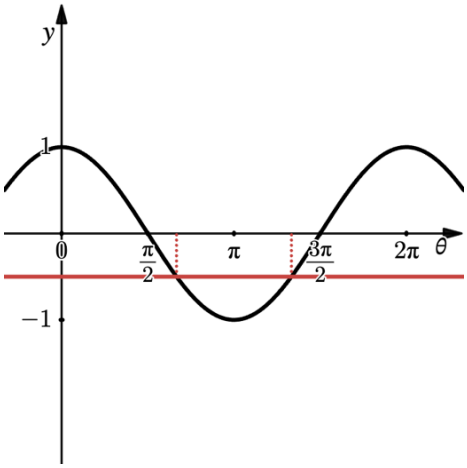
$\theta = \{-2.49, -1.93, -1.24, -0.672\}$ to 3sf.

45 b)

$\cos(7\theta - 4.2) = -0.12$ for $\pi \leq \theta < \frac{3\pi}{2}$.

Find first value between 0 and 360:

$$7\theta - 4.2 = \arccos(-0.12) = 1.6910\ldots$$



Find second value between 0 and 2π: $7\theta - 4.2 = 2\pi - 1.6910\ldots = 4.5920\ldots$.

Apply transformation:

$$7\theta - 4.2 = 1.6910\ldots \Rightarrow \theta = 0.8415\ldots \quad \text{(not in range)}$$

$$7\theta - 4.2 = 4.5920\ldots \Rightarrow \theta = 1.2560\ldots \quad \text{(not in range)}$$

Add period:

$$\theta = 0.8415\ldots + \frac{2\pi}{7} = 1.7391\ldots \quad \text{(not in range)}$$

$$\theta = 1.7391\ldots + \frac{2\pi}{7} = 2.6367\ldots \quad \text{(not in range)}$$

$$\theta = 2.6367\ldots + \frac{2\pi}{7} = 3.5343\ldots \quad \text{(in range)}$$

$$\theta = 3.5243\ldots + \frac{2\pi}{7} = 4.4319\ldots \quad \text{(in range)}$$

$$\theta = 1.2560\ldots + \frac{2\pi}{7} = 2.1536\ldots \quad \text{(not in range)}$$

$$\theta = 2.1536\ldots + \frac{2\pi}{7} = 3.0512\ldots \quad \text{(not in range)}$$

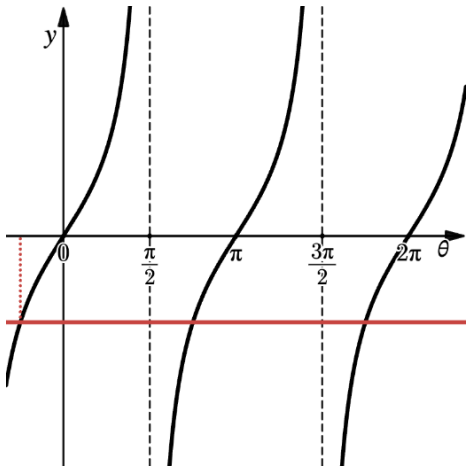
$$\theta = 3.0512\ldots + \frac{2\pi}{7} = 3.9488\ldots \quad \text{(in range)}$$

$\theta = \{3.53, 3.95, 4.43\}$ to 3sf.

45 c)

$\tan(3\theta - 6.4) = -12$ for $2\pi \leq \theta < 4\pi$.

Find first value: $3\theta - 6.4 = \arctan(-12) = -1.4876\ldots$.



Apply transformation: $3\theta - 6.4 = -1.4876\ldots \Rightarrow \theta = 1.6374\ldots$ (not in range).

Add period:

$$\theta = 1.6374\ldots + \frac{\pi}{3} = 2.6846\ldots \quad \text{(not in range)}$$

$$\theta = 2.6846\ldots + \frac{\pi}{3} = 3.7318\ldots \quad \text{(not in range)}$$

$$\theta = 3.7318\ldots + \frac{\pi}{3} = 4.7790\ldots \quad \text{(not in range)}$$

$$\theta = 4.7790\ldots + \frac{\pi}{3} = 5.8262\ldots \quad \text{(not in range)}$$

$$\theta = 5.8262\ldots + \frac{\pi}{3} = 6.8734\ldots \quad \text{(in range)}$$

$$\theta = 6.8734\ldots + \frac{\pi}{3} = 7.9206\ldots \quad \text{(in range)}$$

$$\theta = 7.9206\ldots + \frac{\pi}{3} = 8.9678\ldots \quad \text{(in range)}$$

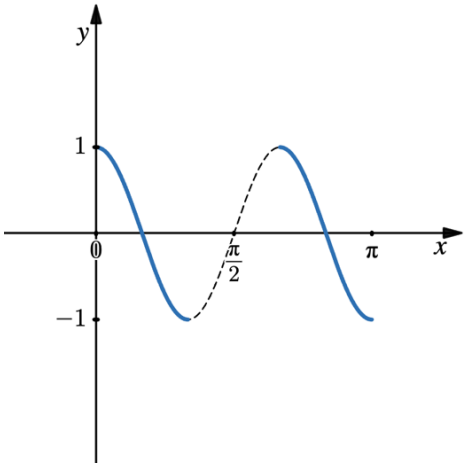
$$\theta = 8.9678\ldots + \frac{\pi}{3} = 10.0150\ldots \quad \text{(in range)}$$

$$\theta = 10.0150\ldots + \frac{\pi}{3} = 11.0622\ldots \quad \text{(in range)}$$

$$\theta = 11.0622\ldots + \frac{\pi}{3} = 12.1094\ldots \quad \text{(in range)}$$

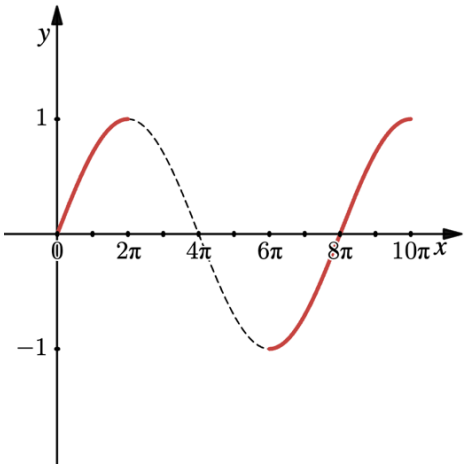
$\theta = \{6.87, 7.92, 8.97, 10.0, 11.1, 12.1\}$ to 3sf.

46)



$0 < x < \frac{\pi}{3}$ or $\frac{2\pi}{3} < x < \pi$.

47)

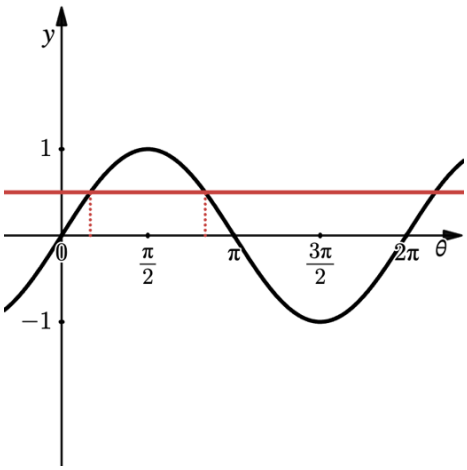


$0 < x < 2\pi$ or $6\pi < x < 10\pi$.

48 a)

$\sin(\frac{\pi}{8} - 2\theta) = \frac{\sqrt{2}}{2}$ for $0 \leq \theta < 2\pi$.

Find first value between 0 and 2π: $\frac{\pi}{8} - 2\theta = \arcsin\frac{\sqrt{2}}{2} = \frac{\pi}{4}$.



Find second value between 0 and 2π: $\frac{\pi}{8} - 2\theta = \pi - \frac{\pi}{4} = \frac{3\pi}{4}$.

Apply transformation:

$$\frac{\pi}{8} - 2\theta = \frac{\pi}{4} \Rightarrow \theta = -\frac{\pi}{16} \quad \text{(not in range)}$$

$$\frac{\pi}{8} - 2\theta = \frac{3\pi}{4} \Rightarrow \theta = -\frac{5\pi}{16} \quad \text{(not in range)}$$

Add period:

$$\theta = -\frac{\pi}{16} + \pi = \frac{15\pi}{16} \quad \text{(in range)}$$

$$\theta = \frac{15\pi}{16} + \pi = \frac{31\pi}{16} \quad \text{(in range)}$$

$$\theta = -\frac{5\pi}{16} + \pi = \frac{11\pi}{16} \quad \text{(in range)}$$

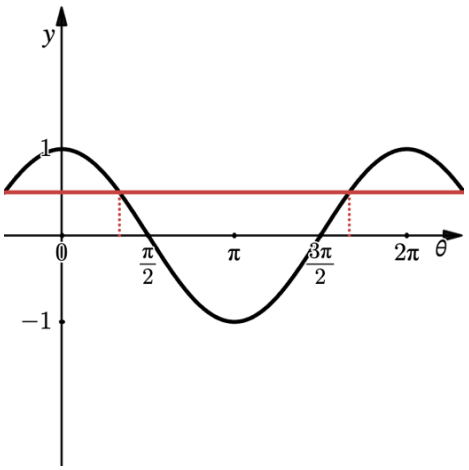
$$\theta = \frac{11\pi}{16} + \pi = \frac{27\pi}{16} \quad \text{(in range)}$$

$\theta = \{\frac{11\pi}{16}, \frac{15\pi}{16}, \frac{27\pi}{16}, \frac{31\pi}{16}\}$.

48 b)

$\cos(\frac{3\pi}{5} - 3\theta) = \frac{\sqrt{3}}{2}$ for $0 \leq \theta < 2\pi$.

Find first value between 0 and 2π: $\frac{3\pi}{5} - 3\theta = \arccos\frac{\sqrt{3}}{2} = \frac{\pi}{6}$.



Find second value between 0 and 2π: $\frac{3\pi}{5} - 3\theta = 2\pi - \frac{\pi}{6} = \frac{11\pi}{6}$.

Apply transformation:

$$\frac{3\pi}{5} - 3\theta = \frac{\pi}{6} \Rightarrow \theta = \frac{13\pi}{90} \quad \text{(in range)}$$

$$\frac{3\pi}{5} - 3\theta = \frac{11\pi}{6} \Rightarrow \theta = -\frac{37\pi}{90} \quad \text{(not in range)}$$

Add period:

$$\theta = \frac{13\pi}{90} + \frac{2\pi}{3} = \frac{73\pi}{90} \quad \text{(in range)}$$

$$\theta = \frac{73\pi}{90} + \frac{2\pi}{3} = \frac{133\pi}{90} \quad \text{(in range)}$$

$$\theta = -\frac{37\pi}{90} + \frac{2\pi}{3} = \frac{23\pi}{90} \quad \text{(in range)}$$

$$\theta = \frac{23\pi}{90} + \frac{2\pi}{3} = \frac{83\pi}{90} \quad \text{(in range)}$$

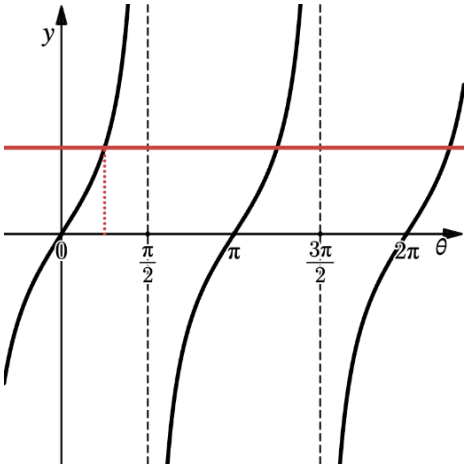
$$\theta = \frac{83\pi}{90} + \frac{2\pi}{3} = \frac{143\pi}{90} \quad \text{(in range)}$$

$\theta = \{\frac{13\pi}{90}, \frac{23\pi}{90}, \frac{73\pi}{90}, \frac{83\pi}{90}, \frac{133\pi}{90}, \frac{143\pi}{90}\}$.

48 c)

$\tan(\frac{3\pi}{10} - 4\theta) = 1$ for $0 \leq \theta < 2\pi$.

Find first value: $\frac{3\pi}{10} - 4\theta = \arctan 1 = \frac{\pi}{4}$.



Apply transformation: $\frac{3\pi}{10} - 4\theta = \frac{\pi}{4} \Rightarrow \theta = \frac{\pi}{80}$ (in range).

Add period:

$$\theta = \frac{\pi}{80} + \frac{\pi}{4} = \frac{21\pi}{80} \quad \text{(in range)}$$

$$\theta = \frac{21\pi}{80} + \frac{\pi}{4} = \frac{41\pi}{80} \quad \text{(in range)}$$

$$\theta = \frac{41\pi}{80} + \frac{\pi}{4} = \frac{61\pi}{80} \quad \text{(in range)}$$

$$\theta = \frac{61\pi}{80} + \frac{\pi}{4} = \frac{81\pi}{80} \quad \text{(in range)}$$

$$\theta = \frac{81\pi}{80} + \frac{\pi}{4} = \frac{101\pi}{80} \quad \text{(in range)}$$

$$\theta = \frac{101\pi}{80} + \frac{\pi}{4} = \frac{121\pi}{80} \quad \text{(in range)}$$

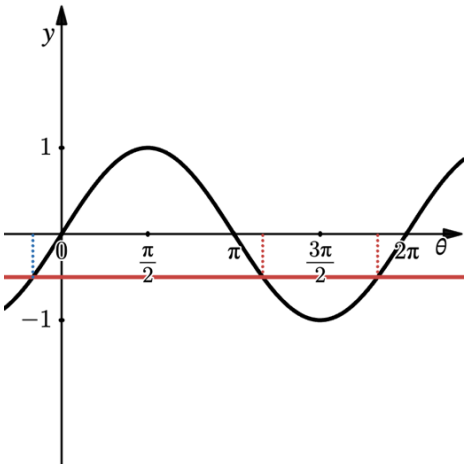
$$\theta = \frac{121\pi}{80} + \frac{\pi}{4} = \frac{141\pi}{80} \quad \text{(in range)}$$

$\theta = \{\frac{\pi}{80}, \frac{21\pi}{80}, \frac{41\pi}{80}, \frac{61\pi}{80}, \frac{81\pi}{80}, \frac{101\pi}{80}, \frac{121\pi}{80}, \frac{141\pi}{80}\}$.

49 a)

$\sin(1.2 - 4\theta) = -0.07$ for $0 \leq \theta < 2$.

Find first value: $1.2 - 4\theta = \arcsin(-0.07) = -0.0700\ldots$.



Find first value between 0 and 2π: $1.2 - 4\theta = \pi + 0.0700\ldots = 3.2116\ldots$.

Find second value between 0 and 2π: $1.2 - 4\theta = 2\pi - 0.0700\ldots = 6.2131\ldots$.

Apply transformation:

$$1.2 - 4\theta = 3.2116\ldots \Rightarrow \theta = -0.5029\ldots \quad \text{(not in range)}$$

$$1.2 - 4\theta = 6.2131\ldots \Rightarrow \theta = -1.2532\ldots \quad \text{(not in range)}$$

Add period:

$$\theta = -0.5029\ldots + \frac{\pi}{2} = 1.0678\ldots \quad \text{(in range)}$$

$$\theta = -1.2532\ldots + \frac{\pi}{2} = 0.3175\ldots \quad \text{(in range)}$$

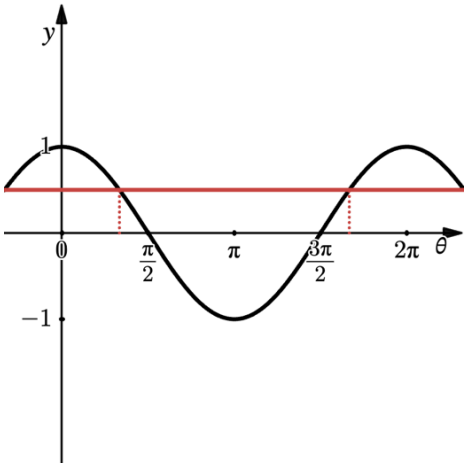
$$\theta = 0.3175\ldots + \frac{\pi}{2} = 1.8883\ldots \quad \text{(in range)}$$

$\theta = \{0.318, 1.07, 1.89\}$ to 3sf.

49 b)

$\cos(2.1 - 5\theta) = 0.93$ for $0 \leq \theta < 1.5$.

Find first value between 0 and 2π: $2.1 - 5\theta = \arccos 0.93 = 0.3763\ldots$.



Find second value between 0 and 2π: $2.1 - 5\theta = 2\pi - 0.3763\ldots = 5.9068\ldots$.

Apply transformation:

$$2.1 - 5\theta = 0.3763\ldots \Rightarrow \theta = 0.3447\ldots \quad \text{(in range)}$$

$$2.1 - 5\theta = 5.9068\ldots \Rightarrow \theta = -0.7613\ldots \quad \text{(not in range)}$$

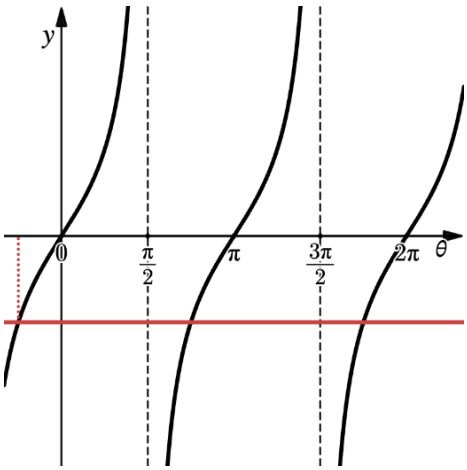
Add period: $\theta = -0.7613\ldots + \frac{2\pi}{5} = 0.4952\ldots$ (in range).

$\theta = \{0.345, 0.495\}$ to 3sf.

49 c)

$\tan(4.3 - 7\theta) = -0.16$ for $0 \leq \theta < 1$.

Find first value: $4.3 - 7\theta = \arctan(-0.16) = -0.1586\ldots$.



Apply transformation: $4.3 - 7\theta = -0.1586\ldots \Rightarrow \theta = 0.6369\ldots$ (in range).

Subtract period: $\theta = 0.6369\ldots - \frac{\pi}{7} = 0.1881\ldots$ (in range).

$\theta = \{0.188, 0.637\}$ to 3sf.

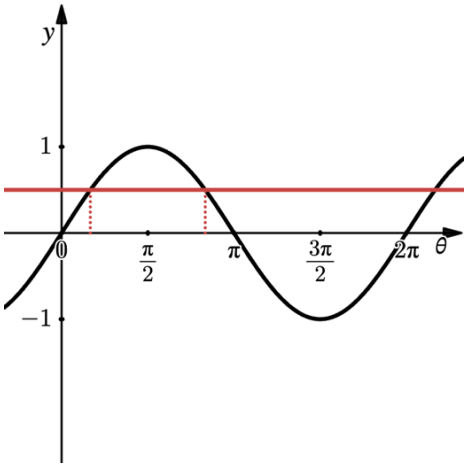
50 a)

$\sin^2 4\theta - \frac{1}{25} = 0$ for $-1.4 \leq \theta < 1.4$.

$$\Rightarrow \sin^2 4\theta = \frac{1}{25} \Rightarrow \sin 4\theta = \frac{1}{5} \text{ or } \sin 4\theta = -\frac{1}{5}$$

First solve $\sin 4\theta = \frac{1}{5}$ for $-1.4 \leq \theta < 1.4$:

Find first value between 0 and 2π: $4\theta = \arcsin \frac{1}{5} = 0.2013\ldots$.



Find second value between 0 and 2π: $4\theta = \pi - 0.2013\ldots = 2.9402\ldots$.

Apply transformation:

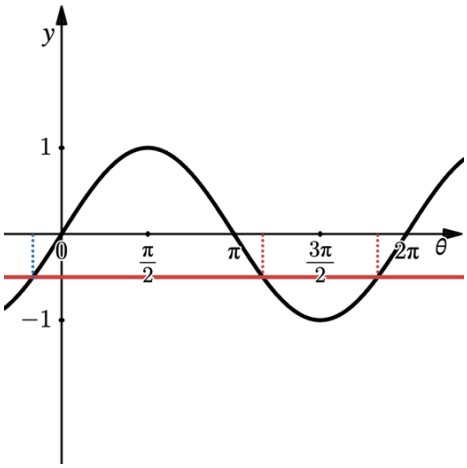
$4\theta = 0.2013\ldots \Rightarrow \theta = 0.0503\ldots$ (in range)

$4\theta = 2.9402\ldots \Rightarrow \theta = 0.7350\ldots$ (in range)

Subtract period: $\theta = 0.7350\ldots - \frac{\pi}{2} = -0.8357\ldots$ (in range).

Now solve $\sin 4\theta = -\frac{1}{5}$ for $-1.4 \leq \theta < 1.4$:

Find first value: $4\theta = \arcsin(-\frac{1}{5}) = -0.2013\ldots$.



Find first value between 0 and 2π: $4\theta = \pi + 0.2013\ldots = 3.3429\ldots$.

Find second value between 0 and 2π: $4\theta = 2\pi - 0.2013\ldots = 6.0818\ldots$.

Apply transformation:

$$4\theta = 3.3429\ldots \Rightarrow \theta = 0.8357\ldots \quad \text{(in range)}$$

$$4\theta = 6.0818\ldots \Rightarrow \theta = 1.5204\ldots \quad \text{(not in range)}$$

Subtract period:

$$\theta = 0.8357\ldots - \frac{\pi}{2} = -0.7350\ldots \quad \text{(in range)}$$

$$\theta = 1.5204\ldots - \frac{\pi}{2} = -0.0503\ldots \quad \text{(in range)}$$

$\theta = \{-0.836, -0.735, -0.0503, 0.0503, 0.735, 0.836\}$ to 3sf.

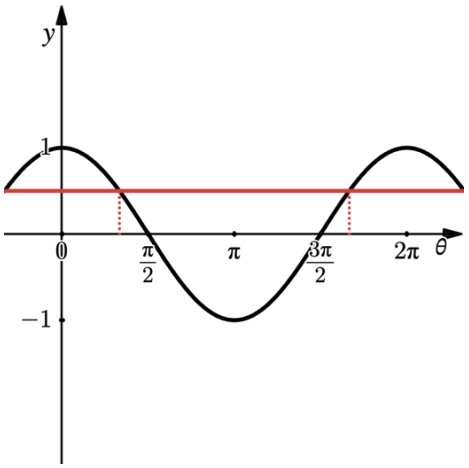
50 b)

$\cos^2 3\theta - \frac{1}{81} = 0$ for $-2 \leq \theta < 2$.

$$\Rightarrow \cos^2 3\theta = \frac{1}{81} \Rightarrow \cos 3\theta = \frac{1}{9} \text{ or } \cos 3\theta = -\frac{1}{9}$$

First solve $\cos 3\theta = \frac{1}{9}$ for $-2 \leq \theta < 2$:

Find first value between 0 and 2π: $3\theta = \arccos \frac{1}{9} = 1.4594\ldots$.



Find second value between 0 and 2π: $3\theta = 2\pi - 1.4594\ldots = 4.8237\ldots$.

Apply transformation:

$$3\theta = 1.4594\ldots \Rightarrow \theta = 0.4864\ldots \quad \text{(in range)}$$

$$3\theta = 4.8237\ldots \Rightarrow \theta = 1.6079\ldots \quad \text{(in range)}$$

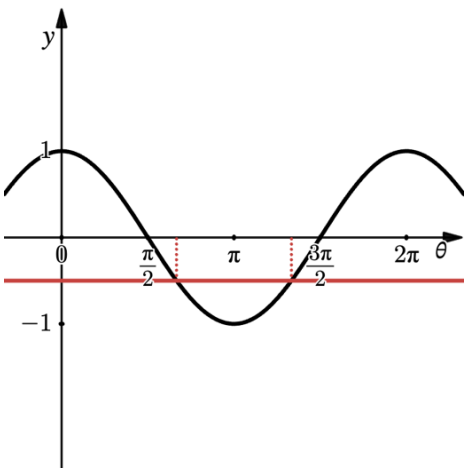
Subtract period:

$$\theta = 0.4864\ldots - \frac{2\pi}{3} = -1.6079\ldots \quad \text{(in range)}$$

$$\theta = 1.6079\ldots - \frac{2\pi}{3} = -0.4864\ldots \quad \text{(in range)}$$

Now solve $\cos 3\theta = -\frac{1}{9}$ for $-2 \leq \theta < 2$:

Find first value between 0 and 2π: $3\theta = \arccos(-\frac{1}{9}) = 1.6821\ldots$.



Find second value between 0 and 2π: $3\theta = 2\pi - 1.6821\ldots = 4.6010\ldots$.

Apply transformation:

$$3\theta = 1.6821\ldots \Rightarrow \theta = 0.5607\ldots \quad \text{(in range)}$$
$$3\theta = 4.6010\ldots \Rightarrow \theta = 1.5336\ldots \quad \text{(in range)}$$

Subtract period:

$$\theta = 0.5607\ldots - \frac{2\pi}{3} = -1.5336\ldots \quad \text{(in range)}$$
$$\theta = 1.5336\ldots - \frac{2\pi}{3} = -0.5607\ldots \quad \text{(in range)}$$

$\theta = \{-1.61, -1.53, -0.561, -0.486, 0.486, 0.561, 1.53, 1.61\}$ to 3sf.

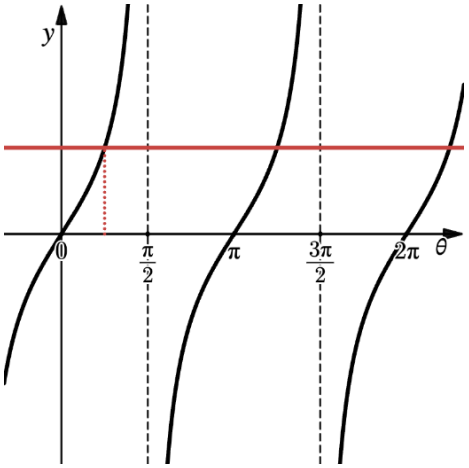
50 c)

$\tan^2 9\theta - 169 = 0$ for $-0.5 \leq \theta < 0.5$.

$$\Rightarrow \tan^2 9\theta = 169 \Rightarrow \tan 9\theta = 13 \text{ or } \tan 9\theta = -13$$

First solve $\tan 9\theta = 13$ for $-0.5 \leq \theta < 0.5$:

Find first value: $9\theta = \arctan 13 = 1.4940\ldots$.

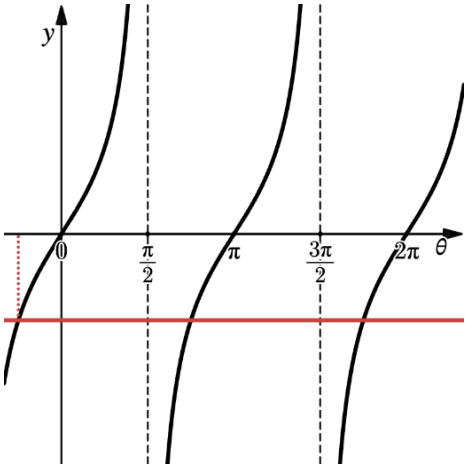


Apply transformation: $9\theta = 1.4940\ldots \Rightarrow \theta = 0.1660\ldots$ (in range).

Subtract period: $\theta = 0.1660\ldots - \frac{\pi}{9} = -0.1830\ldots$ (in range).

Now solve $\tan 9\theta = -13$ for $-0.5 \leq \theta < 0.5$:

Find first value: $9\theta = \arctan(-13) = -1.4940\ldots$.



Apply transformation: $9\theta = -1.4940\ldots \Rightarrow \theta = -0.1660\ldots$ (in range).

Add period: $\theta = -0.1660\ldots + \frac{\pi}{9} = 0.1830\ldots$ (in range).

$\theta = \{-0.183, -0.166, 0.166, 0.183\}$ to 3sf.

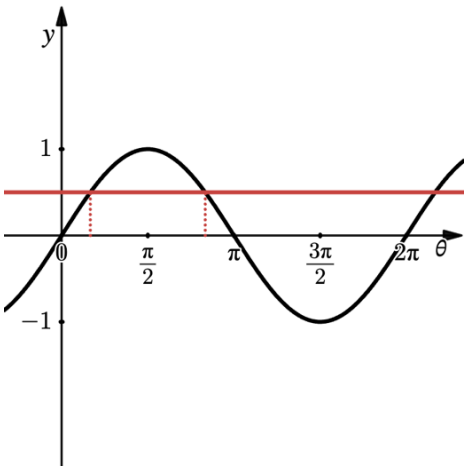
51)

$$A: \ x = -p$$
$$B: \ x = \pi - p$$
$$C: \ x = \frac{5\pi}{3} - p$$
$$D: \ x = p + \frac{2\pi}{3}$$

52 a)

$\sin(\theta - \frac{\pi}{3}) = \sin\frac{\pi}{6}$ for $0 \leq \theta < 2\pi$.

Find first value between 0 and 2π: $\theta - \frac{\pi}{3} = \arcsin(\sin\frac{\pi}{6}) = \frac{\pi}{6}$.



Find second value between 0 and 2π: $\theta - \frac{\pi}{3} = \pi - \frac{\pi}{6} = \frac{5\pi}{6}$.

Apply transformation:

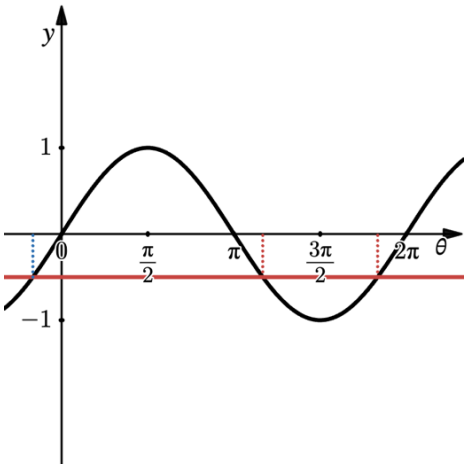
$$\theta - \frac{\pi}{3} = \frac{\pi}{6} \Rightarrow \theta = \frac{\pi}{2} \quad \text{(in range)}$$
$$\theta - \frac{\pi}{3} = \frac{5\pi}{6} \Rightarrow \theta = \frac{7\pi}{6} \quad \text{(in range)}$$

$\theta = \{\frac{\pi}{2}, \frac{7\pi}{6}\}$.

52 b)

$\sin(\theta + \frac{2\pi}{5}) = \sin\frac{\pi}{10}$ for $0 \leq \theta < 2\pi$.

Find first value: $\theta + \frac{2\pi}{5} = \arcsin(\sin\frac{11\pi}{10}) = -\frac{\pi}{10}$.



Find first value between 0 and 2π: $\theta + \frac{2\pi}{5} = \pi + \frac{\pi}{10} = \frac{11\pi}{10}$.

Find second value between 0 and 2π: $\theta + \frac{2\pi}{5} = 2\pi - \frac{\pi}{10} = \frac{19\pi}{10}$.

Apply transformation:

$$\theta + \frac{2\pi}{5} = \frac{11\pi}{10} \Rightarrow \theta = \frac{7\pi}{10} \quad \text{(in range)}$$

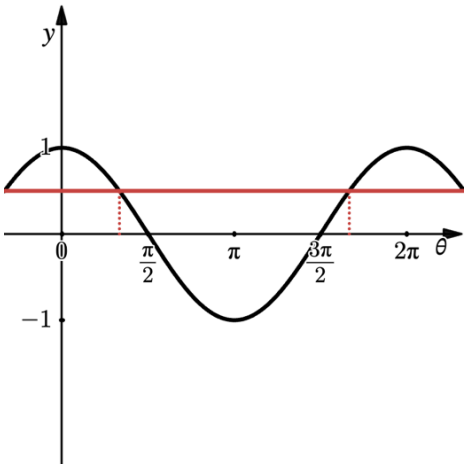
$$\theta + \frac{2\pi}{5} = \frac{19\pi}{10} \Rightarrow \theta = \frac{3\pi}{2} \quad \text{(in range)}$$

$\theta = \{\frac{7\pi}{10}, \frac{3\pi}{2}\}$.

52 c)

$\cos(\theta + \frac{3\pi}{7}) = \cos\frac{4\pi}{3}$ for $0 \leq \theta < 2\pi$.

Find first value between 0 and 2π: $\theta + \frac{3\pi}{7} = \arccos(\cos\frac{4\pi}{3}) = \frac{2\pi}{3}$.



Find second value between 0 and 2π: $\theta + \frac{3\pi}{7} = 2\pi - \frac{2\pi}{3} = \frac{4\pi}{3}$.

Apply transformation:

$$\theta + \frac{3\pi}{7} = \frac{2\pi}{3} \Rightarrow \theta = \frac{5\pi}{21} \quad \text{(in range)}$$

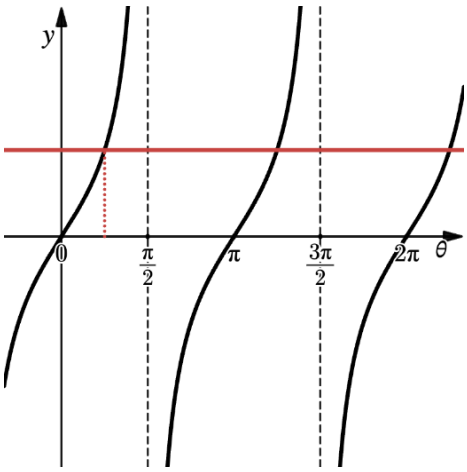
$$\theta + \frac{3\pi}{7} = \frac{4\pi}{3} \Rightarrow \theta = \frac{19\pi}{21} \quad \text{(in range)}$$

$\theta = \{\frac{5\pi}{21}, \frac{19\pi}{21}\}$.

52 d)

$\tan(\theta - \frac{5\pi}{8}) = \tan\frac{\pi}{16}$ for $0 \leq x < 2\pi$.

Find first value: $\theta - \frac{5\pi}{8} = \arctan(\tan\frac{\pi}{16}) = \frac{\pi}{16}$.



Apply transformation: $\theta - \frac{5\pi}{8} = \frac{\pi}{16} \Rightarrow \theta = \frac{11\pi}{16}$ (in range).

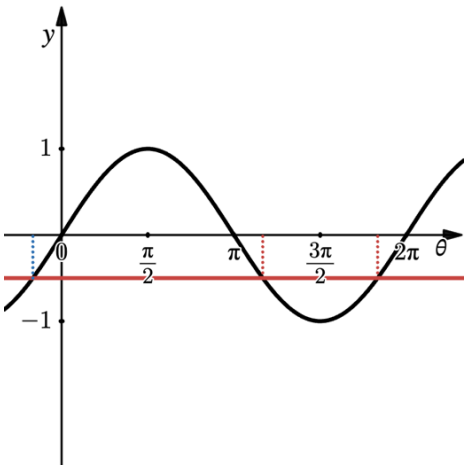
Add period: $\theta = \frac{11\pi}{16} + \pi = \frac{27\pi}{16}$ (in range).

$\theta = \{\frac{11\pi}{16}, \frac{27\pi}{16}\}$.

53 a)

$\sin(\frac{\pi}{3} - \theta) = \sin\frac{46\pi}{9}$ for $0 \leq \theta < 2\pi$.

Find first value: $\frac{\pi}{3} - \theta = \arcsin(\sin\frac{46\pi}{9}) = -\frac{\pi}{9}$.



Find first value between 0 and 2π: $\frac{\pi}{3} - \theta = \pi + \frac{\pi}{9} = \frac{10\pi}{9}$.

Find second value between 0 and 2π: $\frac{\pi}{3} - \theta = 2\pi - \frac{\pi}{9} = \frac{17\pi}{9}$.

Apply transformation:

$$\frac{\pi}{3} - \theta = \frac{10\pi}{9} \Rightarrow \theta = -\frac{7\pi}{9} \quad \text{(not in range)}$$

$$\frac{\pi}{3} - \theta = \frac{17\pi}{9} \Rightarrow \theta = -\frac{14\pi}{9} \quad \text{(not in range)}$$

Add period:

$$\theta = -\frac{7\pi}{9} + 2\pi = \frac{11\pi}{9} \quad \text{(in range)}$$

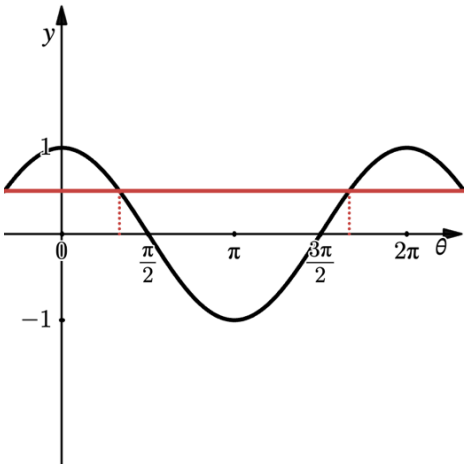
$$\theta = -\frac{14\pi}{9} + 2\pi = \frac{4\pi}{9} \quad \text{(in range)}$$

$\theta = \{\frac{4\pi}{9}, \frac{11\pi}{9}\}$.

53 b)

$\cos(2\theta - \frac{5\pi}{6}) = \cos\frac{11\pi}{12}$ for $0 \leq \theta < 2\pi$.

Find first value between 0 and 2π: $2\theta - \frac{5\pi}{6} = \arccos(\cos\frac{11\pi}{12}) = \frac{11\pi}{12}$.



Find second value between 0 and 2π: $2\theta - \frac{5\pi}{6} = 2\pi - \frac{11\pi}{12} = \frac{13\pi}{12}$.

Apply transformation:

$$2\theta - \frac{5\pi}{6} = \frac{11\pi}{12} \Rightarrow \theta = \frac{7\pi}{8} \quad \text{(in range)}$$

$$2\theta - \frac{5\pi}{6} = \frac{13\pi}{12} \Rightarrow \theta = \frac{23\pi}{24} \quad \text{(in range)}$$

Add period:

$$\theta = \frac{7\pi}{8} + \pi = \frac{15\pi}{8} \quad \text{(in range)}$$

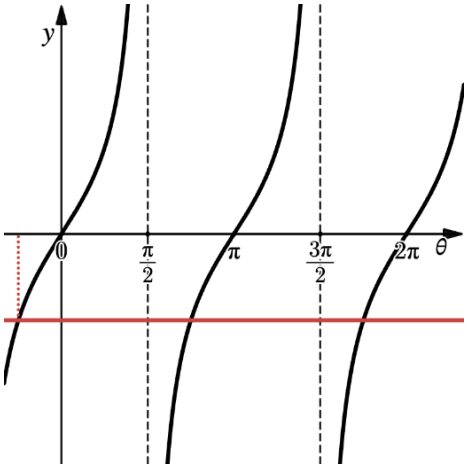
$$\theta = \frac{23\pi}{24} + \pi = \frac{47\pi}{24} \quad \text{(in range)}$$

$\theta = \{\frac{7\pi}{8}, \frac{23\pi}{24}, \frac{15\pi}{8}, \frac{47\pi}{24}\}$.

53 c)

$\tan(\frac{7\pi}{6} - 4\theta) = \tan\frac{8\pi}{3}$ for $0 \leq \theta < 2\pi$.

Find first value: $\frac{7\pi}{6} - 4\theta = \arctan(\tan\frac{8\pi}{3}) = -\frac{\pi}{3}$.



Apply transformation: $\frac{7\pi}{6} - 4\theta = -\frac{\pi}{3} \Rightarrow \theta = \frac{3\pi}{8}$ (in range).

Subtract period: $\theta = \frac{3\pi}{8} - \frac{\pi}{4} = \frac{\pi}{8}$ (in range).

Add period:

$$\theta = \frac{3\pi}{8} + \frac{\pi}{4} = \frac{5\pi}{8} \quad \text{(in range)}$$

$$\theta = \frac{5\pi}{8} + \frac{\pi}{4} = \frac{7\pi}{8} \quad \text{(in range)}$$

$$\theta = \frac{7\pi}{8} + \frac{\pi}{4} = \frac{9\pi}{8} \quad \text{(in range)}$$

$$\theta = \frac{9\pi}{8} + \frac{\pi}{4} = \frac{11\pi}{8} \quad \text{(in range)}$$

$$\theta = \frac{11\pi}{8} + \frac{\pi}{4} = \frac{13\pi}{8} \quad \text{(in range)}$$

$$\theta = \frac{13\pi}{8} + \frac{\pi}{4} = \frac{15\pi}{8} \quad \text{(in range)}$$

$\theta = \{\frac{\pi}{8}, \frac{3\pi}{8}, \frac{5\pi}{8}, \frac{7\pi}{8}, \frac{9\pi}{8}, \frac{11\pi}{8}, \frac{13\pi}{8}, \frac{15\pi}{8}\}$.